Bachelor's Thesis

Domain adapted topic modelling for alt-tech social media

Institute of Artificial Intelligence (AI) in Management
Ludwig-Maximilians-Universität München

Andreea Elena Bodea

Prof. Dr. Stefan Feuerriegel
Academic Advisor: Dominik Bär

Munich, November 11th, 2022

Abstract

Due to some recent violent events like the U.S. Capitol attack and the Pittsburgh synagogue shooting, which were linked to the alt-tech platforms Parler and Gab, the relevance of analyzing alt-tech social media posts has increased. Topic models, particularly neural ones, are able to extract subjects from many domains efficiently. However, these models were never explicitly designed for alt-tech social media, although domain adaptation has been proven to enhance the performance of models. The aim of this thesis is to analyze the performance of domain adapted topic models for alt-tech social media data. Therefore, both classic topic models like LDA and NMF and several variations of the neural topic model BERTopic were implemented, thoroughly evaluated and compared. Domain adaptation was performed by further training SBERT sentence embeddings using the TSDAE and SimCSE techniques. The findings confirm that neural topic models outperform traditional models even when applied to alt-tech social media posts. More significantly, domain adapted neural topic models achieve the best performance for topic extraction. Topic diversity improves when models are domain adapted using data from multiple alt-tech social media domains, while topic coherence is enhanced by further training and testing the models on data from the same platform.

Table of Contents

1 Introduction **2**
1.1 Research Questions . . . 3

2 Theoretical Foundation **5**
2.1 Alt-tech Social Media . . . 5
2.1.1 Parler . . . 5
2.1.2 Gab . . . 6
2.2 Topic Modelling . . . 7
2.2.1 LDA . . . 8
2.2.2 NMF . . . 8
2.2.3 BERTopic . . . 9
2.3 Domain Adaptation . . . 11
2.3.1 Sentence-BERT . . . 11
2.3.2 TSDAE . . . 12
2.3.3 SimCSE . . . 13
2.4 Evaluation of Topic Models . . . 13

3 Methodology **16**
3.1 Datasets . . . 16
3.1.1 Parler Dataset . . . 16
3.1.2 Gab Dataset . . . 17
3.2 Data Preprocessing and Sample Formation . . . 18
3.3 Topic Models . . . 21
3.3.1 LDA and NMF using OCTIS . . . 22
3.3.2 BERTopic Models with SBERT . . . 22
3.3.3 Evaluation . . . 22

4 Results **24**
4.1 Extracted Topics . . . 24
4.2 Word Frequency and Topic Categories . . . 26
4.3 Survey . . . 28
4.4 Quantitative Metrics . . . 29

5 Discussion **30**
5.1 Research Questions Answers . . . 30
5.2 Strengths and Limitations . . . 32

6 Conclusion **34**

A Appendix **35**

B List of Abbreviations **45**

List of Figures

1 NMF 9
2 BERTopic 10
3 SBERT 12
4 Preprocessing pipeline 18
5 Parler sample formation 20
6 Survey evaluation 28

List of Tables

1 Topics 25
2 Word frequency 26
3 Topic categories 27
4 Topic evaluation metrics 29
5 Topics extracted by LDA model 41
6 Topics extracted by NMF model 41
7 Topics extracted by Model 0: BERTopic with pre-trained SBERT 42
8 Topics extracted by Model 1: BERTopic with SBERT trained using TSDAE on Parler data 43
9 Topics extracted by Model 2: BERTopic with SBERT trained using SimCSE on Parler data 43
10 Topics extracted by Model 3: BERTopic with SBERT trained using TSDAE on Gab data 44
11 Topics extracted by Model 4: BERTopic with SBERT trained using TSDAE on Parler and Gab data 44

1 Introduction

Topic modelling algorithms are some of the most used text mining tools in the Natural Language Processing (NLP) field. They extract latent variables, called topics, from collections of documents, generating efficient representations of a corpus.

Topic models help solve numerous NLP-related problems and are applied to diverse domains. Some examples of tasks where these techniques find applicability are text summarization and categorization, question answering, sentiment analysis, understanding stance and polarization, and even machine translation (Boyd-Graber et al., 2017). Apart from the many areas of implementation in the NLP field, these models are also used for extracting topics from a large variety of domains. For instance, topic modelling algorithms can be applied to scientific data (ex. research papers), historical documents (ex. newspapers and historical records), but also to short-text data like social media posts (Boyd-Graber et al., 2017).

Over the past two decades, multiple topic models have been proposed. Among the classic techniques, Latent Dirichlet Allocation (LDA) (Blei et al., 2003) and Non-Negative Matrix Factorization (NMF) (Lee & Seung, 2000) stand out, because they usually deliver coherent and intelligible results. Yet, with the recent development of neural networks and the transformer architecture, neural topic models (NTMs) like BERTopic were introduced (Grootendorst, 2022b). According to different studies, NTMs outperform traditional techniques on a variety of topic modelling domains (Zhao et al., 2021) (Bennett et al., 2021). However, these novel algorithms are still intensively researched and evaluated. Furthermore, the assessment and comparison of topic models is not a trivial task, because they rely on unsupervised learning and a perfect solution, often referred to as "ground truth", is not available.

NTMs are based on large language models, which are usually pre-trained on huge corpora of common language texts. For example, the original BERT model (Devlin et al., 2018), which stands for Bidirectional Encoder Representations from Transformers, has been pre-trained on the BooksCorpus dataset consisting of 800 million words and on the English Wikipedia corpus which contains around 2,500 million words. Although these language models work very well on standard language, when dealing with text related to a specific domain they are not able to understand niche language accurately. Researchers have tried to improve the results of the models for particular fields. This process is called domain adaptation and is described as the further training, also referred to as fine-tuning, of a pre-trained model on a new dataset, aiming to achieve predictions adjusted to the field or task. Examples of domain adapted language models based on BERT are ClimateBERT for climate-related text (Webersinke et al., 2021), FinBERT for the finance field (Araci, 2019), and BERTweet, a model pre-trained for English tweets (Nguyen et al., 2020).

Research already showed that domain adapted BERT models deliver more field-specific results in comparison with the pre-trained BERT. However, the accuracy of domain adapted models is still not ideal, which means they can be further refined. One type of insightful text commonly analyzed is user-generated textual content like comments, reviews, and short-text messages posted on social media platforms. However, social media posts provide limited contextual information and they are usually sparse, noisy and ambiguous. Also, they contain abbreviations and slang, which can not be found in lit-

erature or academic writing. In order to address these characteristics properly, several topic modelling techniques were specially designed and tested for short-text data (Qiang et al., 2020) (Albalawi et al., 2020) (Yin & Wang, 2014) (Cheng et al., 2014) (Li et al., 2016) (Shi et al., 2018). Due to some recent events like the 2021 United States Capitol attack and the Pittsburgh synagogue shooting, which were proven to be related to users of certain alternative social media networks (McIlroy-Young & Anderson, 2019), researchers began analyzing the language and user base of alt-tech platforms. Some of the most well-known among these are Parler, Gab, and Rumble (Lewis, 2020). Several studies found that hate speech is prevalent on alt-tech platforms, and extremists and conspiracy theorists are active to a large extent (Bodea, 2022b) (Aliapoulios et al., 2021a) (Zannettou et al., 2018b). Because the language commonly used by the people posting on these social media platforms is distinctive, not even topic models that were adapted for textual posts are able to perform adequately. This research aims to address this gap and presents a rigorous examination of different topic models, training techniques and data sources for domain adaptation applied to the alt-tech social media field.

1.1 Research Questions

Three research questions have been defined:

1. Can neural topic models improve topic discovery in comparison with traditional topic models for alt-tech social media data?
2. Do domain adapted neural topic models enhance topic discovery in comparison with pre-trained neural topic models for alt-tech social media data?
3. How does the source of the data used for training domain adapted neural topic models influence the performance of the models?

The first research question investigates whether or not it makes sense to apply novel neural topic models to the specific domain of alt-tech social media posts. Are the advantages of transformer-based algorithms like the BERTopic model, which relies on pre-trained sentence embeddings, sufficient to outperform the results of traditional models like LDA and NMF? The second question analyzes two different domain adaptation techniques for neural topic models, more specifically for BERTopic, which uses SBERT (Reimers & Gurevych, 2019) as the sentence embedding method. These two techniques, Transformer-based Sequential Denoising Auto-Encoder (TSDAE) (Wang et al., 2021) and Simple Contrastive Learning of Sentence Embeddings (SimCSE) (Gao et al., 2021), although based on different mechanisms, typically enhance the performance of the models. This research evaluates whether they also improve topic discovery for the alt-tech social media domain. The performances of a BERTopic model trained using TSDAE and a BERTopic model trained using SimCSE are compared with the one of a classic BERTopic algorithm based on pre-trained sentence embeddings. The third question looks at the differences in the performance of neural topic models when these are domain adapted using alt-tech social media posts coming from several sources. To be able to compare the models, all of them are tested on data gathered from the same platform. Does the performance improve when training the model on data from the social media platform it will be tested on or is it more

efficient to choose another source? Is training the topic model on posts from both the targeted social media platform and from another platform beneficial for the performance?

In order to answer these questions impartially, this research aims to examine the performance of different topic modelling algorithms on a large dataset of Parler posts and comments from the beginning of November 2020 until the platform's shutdown in January 2021. The samples for training the models are generated from both Parler posts and Gab posts. The final goal is to observe which model enhances the performance the most for the task of topic extraction on alt-tech social media data.

2 Theoretical Foundation

For an in-depth understanding of the research topic of this thesis, defining alt-tech social media, topic modelling, domain adaptation and evaluation metrics for topic models is required.

2.1 Alt-tech Social Media

Alt-tech social media platforms are those social networking platforms that are appealing to people with extremist, alt-right, or far-right opinions. These networks have indulgent content moderation policies and claim they defend free speech and individual liberty (*Alt-tech*, 2022). Alt-tech platforms have generally the same functionality as more popular social networks, allowing users to display their viewpoints, interact with content posted by other users, and share news. The user base of these platforms increased when well-known social media websites like Twitter and Facebook began to impose stricter rules regarding the content allowed to post (Aliapoulios et al., 2021a) (Zannettou et al., 2018b). Users that failed to comply with the new regulations, faced restrictions on activity and got banned. Alt-tech platforms mainly attract people that can not express their opinion on mainstream platforms because of the limits imposed.

According to the number of users and posts, two of the most popular alt-tech social media platform are Parler and Gab (Lewis, 2020).

2.1.1 Parler

With its name making reference to the French verb "to speak", Parler is an American online social network founded in August 2018. The platform promotes itself as "The premier global free speech platform." (*Parler*, 2022) Focusing especially on free speech, Parler is one of the leading alternatives to large social media platforms such as Twitter and Facebook, which restrict the content posted.

Both Prabhu et al. (2021) and M. Otala et al. (2021) compared Parler to Twitter. First, they inspected the functionalities provided by the two platforms and found that they are very similar, even if the functions have different names. Users of Parler are allowed to publish texts (called "parlays") of up to 1,000 characters and can respond to previous messages. In addition, Parler provides a reposting feature called "echos" which are the same as the retweets on Twitter. Parler users can also express their appreciation or disagreement with other people's posts, using the "upvote" and "downvote" functions, similar to "likes" on Twitter. Apart from the functionalities, the researchers also explored the rules imposed by the two platforms, but this time found substantial differences. While on Twitter there is an exhaustive set of limitations that users must respect, on Parler there are only two rules stipulated under "Community guidelines".

As described in a study by Bodea (2022b), the most recent version of the "Community Guidelines" was amended on February 11, 2021 (*Parler Community Guidelines*, 2021). The document explains that Parler aims not to delete posts or exclude users except in extreme cases, leaving the freedom to filter what one desires to see to each person active on the platform. For this, Parler implemented mute and block functions, which can be applied to specific users or comments containing terms of choice. Yet, the platform limits

content according to two different criteria. Their first rule refers to the exclusion of posts related to crimes, civil torts, intellectual property theft, unlawful acts, and child sexual abuse, and of content posted by or on behalf of terrorist organizations. The second rule forbids spam posts and the use of bots.

Parler was linked to the United States Capitol attack that took place in January 2021 (Bodea, 2022b). Trump supporters were detailing the plan for the attack and posted videos and photos with rioters on the platform (MacGillis, 2021). Parler admitted that some posts were referring to the assault on the Capitol building, but it defended itself by declaring that its "mission has always been to offer a social platform in the spirit of the First Amendment of the United States Constitution, one which empowers the individual to think and share freely" (*Parler Community Guidelines*, 2021).

2.1.2 Gab

Created in August 2016, Gab is another alt-tech social media platform, popular among users banned or suspended from classic social networks. It advertises itself as "A social network that champions free speech, individual liberty and the free flow of information online. All are welcome." (*Gab*, 2022) The platform has the same social networking functions as well-known social media websites. Users can send 300-character long messages, called "gabs", to their followers. These can be up or down-voted and include hashtags or mentions. Gabs can be reposted, quoted, and used as replies to other gabs. Pornographic and obscene content is allowed as long as users mark it with "Not-Safe-For-Work". Aiming to protect free speech above anything else, Gab has attracted alt-right users and conspiracy theorists. Not only because of the high volume of hate speech and racism, but also as a result of non-compliance to pornographic content guidelines both Google and Apple rejected Gab's mobile apps from their stores in 2017, but only for a limited period of time. Zannettou et al. (2018b) gathered a large number of posts made on the platform between August 2016 and January 2018 and analyzed them thoroughly. They confirmed that Gab is mostly used for sharing and discussing global political events or conspiracy theories. They also proved that Gab users respond strongly to news about white nationalism and Donald Trump. Many users appeared to be religious, conservative Americans that posted false or controversial information on the platform.

Mathew et al. (2019) analyzed how hate speech, a serious problem in online social media, especially for alt-tech platforms, spreads across Gab. Their main finding reveals that posts made by hateful users tend to spread faster, farther and reach a larger public than the ones posted by normal users. Apart from that, people using hateful comments are connected more densely with others similar to them.

Researching alt-tech social media platforms is relevant because some of their users took part in violent actions. In October 2018, a person active on Gab posted anti-Semitic comments on the platform, entered the Pittsburgh synagogue that he made references to in his posts, and fatally shot 18 people. McIlroy-Young & Anderson (2019) looked at the evolution of Gab until the deadly attack, exploring the language used and the topics discussed over time. They showed that the most common words are related to politics and aligned with alt-right views. Also, the term "jew" seemed to be surrounded by offensive language and ethnic slurs. However, foreseeing violence by tracking hateful comments on the platforms is not straightforward. The shooters' posts on Gab were consistently

including anti-Semitic beliefs, but hundreds of other users were even more extreme.

2.2 Topic Modelling

Another term that needs clarification is topic modelling. It refers to the NLP task which deals with the extraction of subjects from text documents.

Zhao et al. (2021) analyzed recent advances in the topic modelling field. The researchers define the purpose of a topic model as the discovery of "a set of latent topics, each of which describes an interpretable semantic concept". They made a clear distinction between Bayesian probabilistic topic models (BPTMs) and neural topic models (NTMs) and recommend the usage of the latter.

Until recently, BPTMs, especially the Latent Dirichlet Allocation (LDA) model, have been the algorithms delivering the best results. LDA remains the most known topic model and is consistently used as a baseline comparison against other topic models. It follows the principles of an earlier algorithm called Probabilistic Latent Semantic Index, which works by using singular value decomposition on a term frequency matrix.

In contrast to LDA, NMF is a decompositional, non-probabilistic algorithm relying on matrix factorization. It belongs to the group of linear-algebraic algorithms. NMF works on TF-IDF (Term Frequency – Inverse Document Frequency) transformed data by breaking down a matrix into two lower-ranking matrices.

However, with the improvement of deep neural networks, topic modelling advanced and NTMs are the state-of-the-art algorithms nowadays. The development of these algorithms began when Vaswani et al. (2017) introduced the transformer architecture. Considerably improving the use of the previous attention mechanism, this architecture represented a substantial advance in the machine learning field. After only one year, (Devlin et al., 2018) presented the Bidirectional Encoder Representations from Transformers (BERT) model. This new language model was stacking encoders on top of each other using a bidirectional transformer architecture and represented a real breakthrough for a majority of NLP tasks. Word or sentence embeddings trained from neural networks are also used directly to find topic models. In this case, the word embeddings act as the neural network document representation. BERTopic, for example, applies clustering methods to SBERT embeddings to generate topics (Grootendorst, 2022b).

Not only did the algorithms change, but also the way of training. The paradigm of transfer learning has been applied to NLP. It is not necessary anymore to train a model from scratch, which usually takes a lot of time and requires high computational power. Better results have been achieved when using an already pre-trained model and fine-tuning it for a specific task.

The experiments of this thesis include two of the classic topic models, LDA and NMF, and several variations of a neural topic model, namely BERTopic with sentence embedding trained using different approaches. Although some studies proved that neural topic models outperform traditional topic models on a variety of metrics (Zhao et al., 2021), (Bennett et al., 2021), LDA and NMF still represent a valuable baseline Doan & Hoang (2021). The upcoming sections examine the algorithms and training techniques in depth.

2.2.1 LDA

The most well-known topic modelling algorithm, Latent Dirichlet Allocation (LDA) (Blei et al., 2003) is a model that uses Bayesian probability to generate topics. As for the other algorithms implemented for this thesis, specifying the number of topics before training is required. While each topic follows a multinomial distribution of all words in the corpus, the distribution of all topics is defined as a Dirichlet distribution, a non-binary distribution for probabilities. As explained by Hadiat (2022), the steps that LDA goes through to generate topics, i.e find the probability of each word to belong to a topic, are the following:

1. Choose a number of topics K
2. Randomly initialize each topic to a document (document-topic distribution)
3. For each document, randomly assign each word to a topic (topic-word distribution)
4. For each word W in each document D compute:
   - $p(T|D)$: Proportion of words in document D that are assigned to topic T. It evaluates how many words belong to the topic T for a given document D excluding the current word.
   - $p(W|T)$: Proportion of assignments to topic T over all documents that come from the word W. It evaluates how many documents are in topic T because of word W. LDA represents documents as a mixture of topics. Similarly, a topic is a mixture of words. If a word has high probability of being in a topic, all the documents having the word W will be more strongly associated with T as well.
5. Update the probability for the word W to belong to topic T' as $p(T'|D) \cdot p(W|T')$) considering all other words and their topic assignments.

The last step is repeated until the topic assignments do not change further. The proportion of topics for each document is then determined from these topic assignments.

The model assumes that each document is just a collection of words, commonly called "bag of words". Thus, the order and the grammatical role of the words are not considered.

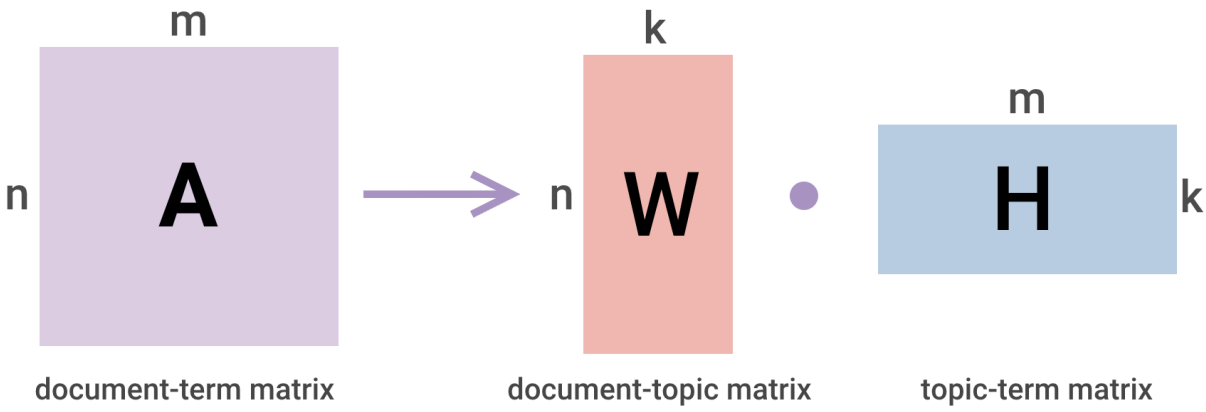
2.2.2 NMF

In contrast to LDA, NMF is a decompositional, non-probabilistic algorithm, which belongs to the group of linear-algebraic algorithms. It stands for "Non-negative Matrix-Factorization", a Latent Semantic Analysis method used to decompose TF-IDF transformed data. As shown in Figure 1 the document-term matrix A with n rows and m columns is divided into a product of two smaller matrices — the document-topic matrix H with n rows and k columns, and the topic-term matrix W with k rows and m columns. Each matrix is populated with unnormalized probabilities, so either zero or positive values. The number of topics k is set beforehand and is required to be equal to or less than the smallest of n and m. The matrix W is generally called the dictionary or basis matrix,

and H is known as expansion or coefficient matrix. TF-IDF is used as a measure to evaluate the importance of a word in a collection of documents. The values of W and H will be modified at every repetition, where the former contains the basis vectors, and the latter contains the corresponding weights. The matrices are determined by minimization of the Frobenius norm carried out by the iterative search process. The solution, i.e. matrices W and H, is not unique.

For better results, the input data should be preprocessed before being forwarded to both NMF and LDA. Required steps include lowercasing, stopword removal, lemmatizing or stemming, as well as punctuation and number removal, which are classic techniques used in an NLP pipeline. Similar to LDA, NMF ignores both syntactic and semantic information. This can lead to incorrect interpretations of words with multiple meanings. Moreover, sentences whose word order contributes to the overall understanding of the context could be misunderstood.

Figure 1: NMF



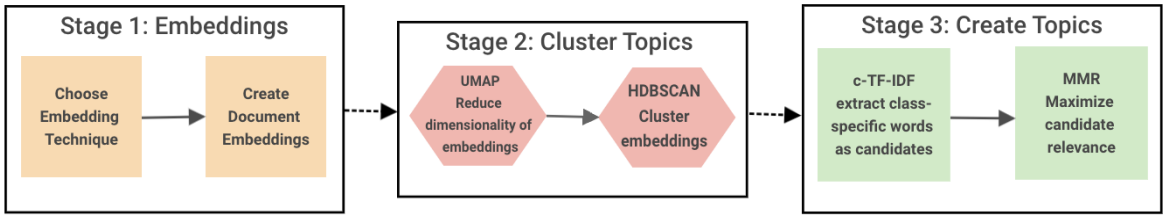
2.2.3 BERTopic

Developed by Grootendorst (2022b), BERTopic is a "topic modelling technique that leverages transformers and c-TF-IDF to create dense clusters allowing for easily interpretable topics whilst keeping important words in the topic descriptions". More exactly, BERTopic utilizes the Bi-directional Encoder Representation Transformer (BERT) model (Devlin et al., 2018) to extract topics from collections of documents. As explained by Hadiat (2022), the algorithm applies multiple steps on a corpus of texts, steps that were grouped into three stages and represented in Figure 2. First, the algorithm transforms documents into embedding representations. Second, it reduces the dimensionality of the embeddings and clusters them. Third, it extracts the topics using a modified version of TF-IDF.

In the first stage, BERTopic transforms documents into embeddings using a specific version of BERT called Sentence-BERT (SBERT) (Reimers & Gurevych, 2019). The purpose of this process is to map the documents to a vector space of real numbers. SBERT uses sentences instead of words for document embeddings which enables the model to capture the meaning based on the context. Several sentence embedding models were created and tested, but SBERT "all-MiniLM-L6-v2" model for the English language

has achieved the most competitive performance for several tasks. However, if a specific task is solved more efficiently by using another embedding technique, the algorithm allows choosing whatever model is more suitable. The modular architecture of BERTopic is one of the significant advantages of the algorithm.

Figure 2: BERTopic



The second stage of BERTopic is clustering and requires two steps. First, the dimensions of the embedding are reduced. SBERT maps sentences to a 384 dimensional dense vector space, which leads to long distances between points, causing difficulty for nearest-neighbor problems. BERTopic uses Uniform Manifold Approximation and Projection (UMAP) for dimensionality reduction (McInnes et al., 2018). Other possible dimensionality reduction techniques, like PCA or t-distributed stochastic neighbor (t-SNE) were considered, but choosing UMAP offers some clear advantages. Some of them are the high speed of computation compared to other algorithms and better preservation of the local and global structure of the data. However, due to the stochastic nature of UMAP, which makes use of randomness, even multiple executions of the exact same algorithm produce slightly different results. The variations are usually small, but hard to avoid without reducing the performance. UMAP works by constructing a high dimensional representation of the data and then approximates it with a lower dimensional graph, which achieves the same representation performance while being more efficient. The UMAP-reduced embeddings are then clustered using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) (McInnes et al., 2017). HDBSCAN is a fast and robust algorithm returning meaningful clusters of varying densities and allowing the identification of outliers. The algorithm uses density to cluster data points together and measures it using a mutual reachability distance. This ensures that dense points remain at the same distance from each other while spare points are pushed away. The algorithm then clusters groups together using a density threshold, so that a group of points is only considered a cluster if its density is above a certain value. HDBSCAN is the hierarchical version of DBSCAN. The algorithm splits clusters depending on local density because global thresholds can yield poor-quality clusters due to varying densities.

In the last stage, representations of topics are created. The first step before the clusters can be used as topics is to generate the topic-word distribution. BERTopic uses a modified term-frequency inverse-document-frequency (TF-IDF) function to evaluate how relevant a word is to a document or cluster of documents. TF-IDF itself is a modified term-count matrix and is defined as:

$$W_{t,d} = tf(t,d) \cdot \log(\frac{N}{df_t})$$

Term frequency $tf(t,d)$ is a matrix of counts of all terms that occur in each document. The inverse document frequency $\log(\frac{N}{df_t})$ measures how much a given term provides to a document by calculating the logarithm of a terms count in all documents. The IDF function ensures that rare terms are not overpowered by frequent terms. BERTopic applies TF-IDF to the clusters of documents to generate the topics. Each cluster is considered a document, not a set of documents, and the results of the algorithm are, therefore, the importance scores for words within a cluster. The most important words in a cluster are considered the descriptions of the topics.

2.3 Domain Adaptation

The next concept that requires a definition is domain adaptation. Training techniques based on deep learning can achieve impressive results, outperforming traditional methods by large margins. Language models are usually trained using deep learning approaches on large amounts of textual, unlabelled data. In this initial training phase, referred to as pre-training, the model learns representations of words and patterns of common language. The most well-known language model is BERT (Devlin et al., 2018). The model has been pre-trained on the BooksCorpus dataset consisting of 800 million words and on the English Wikipedia corpus consisting of around 2,500 million words. BERT, as well as other language models, can be either used directly as a pre-trained model or it can be trained even more. This process of further training a model, sometimes also called fine-tuning, is known in NLP as domain adaptation if it involves data from a specific field. Multiple studies have shown that further training a language model improves the performance tremendously for tasks particular to a field because the model is able to adapt better to domain-specific information (Gururangan et al., 2020). Examples of language models based on BERT that were adapted to a specific domain are ClimateBERT (Webersinke et al., 2021), FinBERT (Araci, 2019) and BERTweet (Nguyen et al., 2020). Depending on the type of data available and the downstream task, fine-tuning can be performed using supervised or unsupervised methods.

2.3.1 Sentence-BERT

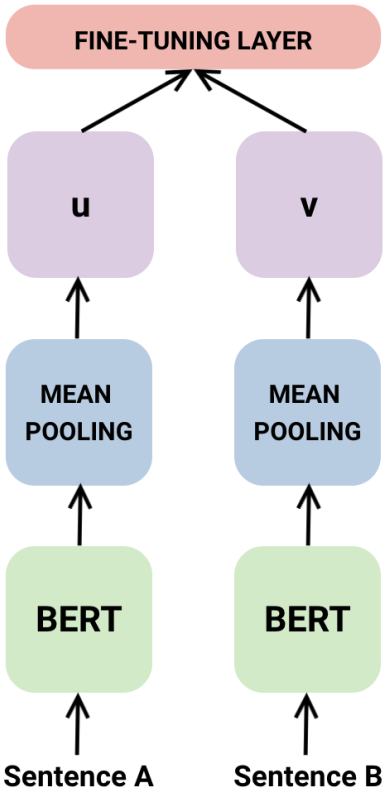
Language models like BERT are adopted for a wide range of NLP tasks and in many different forms. Although outperforming previous models for tasks like question answering and language inference, the construction of the original BERT is unsuitable for semantic similarity search or unsupervised learning (Reimers & Gurevych, 2019). To solve these tasks researchers developed new models based on the transformer architecture of BERT. By default, topic modelling relies on unsupervised learning and many state-of-the-art algorithms that extract topics, including BERTopic, use clustering. Therefore, a variation of BERT called Sentence-BERT (Reimers & Gurevych, 2019) was adapted for several neural topic modelling algorithms. Sentence-BERT (SBERT) is one of the most popular and powerful sentence embedding techniques that uses BERT as a transformer-based model (Reimers & Gurevych, 2019).

Because computers can only work with numeric inputs, text needs to be converted to numbers. Thus, words and sentences are embedded into real-valued vectors that aim to capture the meaning of the word or sentence. In the vector space, words or sentences that are close to one another will have similar meanings.

As a sentence embedding technique, SBERT combines four key concepts: attention, transformers, BERT and siamese networks. Illustrated in Figure 3, the architecture of SBERT is based on a twin network, which allows the simultaneous processing of two sentences. In order to generate their embeddings, each of the two sentences is passed through a BERT model and then through a pooling layer. The pooling layer enables the creation of a fixed-size representation for input sentences of varying lengths. The pooling strategy that performs the best according to the experiments of the researchers is the mean of all output vectors. The calculated sentence embeddings are then forwarded to a fine-tuning layer, which depends on the task that needs to be solved.

The scarcity of labeled alt-tech social media data required searching for unsupervised domain-adaptation methods. For the case study of this thesis, TSDAE and SimCSE, two unsupervised fine-tuning techniques for adapting SBERT, have been chosen.

Figure 3: SBERT



2.3.2 TSDAE

Transformer-based Sequential Denoising Auto-Encoder (TSDAE) (Wang et al., 2021) takes a pre-trained transformer-based model and fine-tunes it using a sequential denoising auto-encoder. The fine-tuning process is done exclusively with sentences from domain-specific data without any labels. The experiments conducted by Wang et al. (2021) showed that the resulting model computes better sentence and paragraph level embeddings compared to the pre-trained version of the input model. During the unsupervised training process, TSDAE first encodes the sentences containing noise into a fixed-length sentence embedding vector. Then, it uses a decoder to reconstruct the original sentence using the generated sentence embedding. The training objective becomes reconstructing previously corrupted input. The idea of TSDAE is similar to one of the tasks the BERT model has been trained on, called Masked Language Modelling (MLM). Both introduce noise to force the encoder to capture the complete meaning of the sentence. More specifically, TSDAE introduces noise by removing some input words randomly. Additionally, the attention from the decoder has limited access to the output from the encoder, namely only the sentence vector and not the token vectors This also puts pressure on the model to create meaningful sentence embeddings. After the training process, the encoder is used to generate the final sentence embedding.

Interestingly, Wang et al. (2021) experimented with several training sample sizes and concluded that TSDAE outperforms previous unsupervised learning methods even when trained with only 1,000 unlabeled sentences. They also noticed that more than 100,000 sentences at training do not further improve performance. This means that less than 100,000 sentences as unlabeled training data are needed to tune pre-trained transformers to a specific domain. TSDAE currently achieves state-of-the-art results compared to other unsupervised fine-tuning techniques.

2.3.3 SimCSE

SimCSE is "a simple contrastive learning framework" for sentence embeddings introduced by Gao et al. (2021). Contrastive learning is a machine learning technique used to determine the general features of unsupervised data by training a model with similar and dissimilar instances. The researchers propose both a supervised and an unsupervised approach for training embeddings at sentence level using SimCSE.

The unsupervised method uses standard dropout as noise and tries to predict a sentence taken as input in a contrastive objective. It requires positive and negative pairs of input sentences for training. Dropout works by applying a mask to a network according to the dropout rate. These process is similar to sampling a sub-network of the initial network. When given a set of sentences, the same sentence is used twice as input and will be represented each time as a slightly different embedding. Then, the two embeddings are used as the positive pair for the training objective and other sentences are regarded as the negative pair. Using dropout is necessary and works as minimal data augmentation. The researchers compared the performance of several augmentation techniques like standard dropout, crop, word deletion, synonym replacement, and masked language modelling and found that the dropout operation achieves the best results. Also, they demonstrate that the training objective of the unsupervised SimCSE is better than the traditional next sentence prediction task, used for training the BERT model. Although a simple, straightforward method, the performance of SimCSE is comparable to previous supervised alternatives.

2.4 Evaluation of Topic Models

The evaluation of topic models is the process of assessing the results of a topic modelling technique. It aims to answer how well the algorithm is performing and whether or not the topics extracted are understandable, coherent, diverse and easy to interpret by humans. However, because of the unsupervised nature of topic models and the lack of a "ground truth" solution, evaluating the models is not an effortless task. Some of the acknowledged ways to assess their performance are:

- observation-based evaluation,
- interpretation-based evaluation,
- metric-based evaluation.

While the first two types of evaluation rely on human judgment, the last one uses mathematical formulas.

The observation-based evaluation method examines the words extracted by the model for each topic. Topic modelling does not provide guidance on the meaning of the topics, so labeling them requires human interpretation. Visual illustrations like tables, graphics, wordclouds or termites (Chuang et al., 2012) are used by most of the studies on topic modelling (Egger & Yu, 2022) (Mifrah & Benlahmar, 2022).

Interpretation-based approaches are considered a "gold standard" for evaluating topic models because they rely solely on human judgment. Although they produce more reliable results, they also require much more effort. Chang et al. (2009) developed the first interpretation-based human evaluation method for topic models. They use people's judgment to measure semantic meaning in inferred topics by presenting them two tasks to solve, called "word intrusion" and "topic intrusion". The "word intrusion" task aims at evaluating whether a topic has human-identifiable semantic coherence. Participants are shown a list of five words from a topic extracted by a model. In each list, a sixth word from a different topic is added. The subjects are assigned with finding the word that does not belong with the others, i.e., the intruder. When the intruder word is consistently identified, then the topic extracted by the model is considered coherent. If the five words extracted by the algorithm are ambiguous, the subjects usually choose an intruder randomly, which means that the topic has poor coherence. The second task, "topic intrusion" tests whether the association between a document and a topic makes sense. Individuals are provided with a document's title, a paragraph from the document and four topics. Three of them are the highest probability topics assigned to that document and one is the intruder, a topic randomly selected from the model's other low-probability topics. Humans are then asked to identify which topic was not associated with the document by the model. Interpretation-based human evaluation is used by high-quality research studies like Toetzke et al. (2022).

The last type of evaluation is based on quantitative metrics calculations, which can be more easily standardized and automated. One of the first intrinsic methods was perplexity, presented by Wallach et al. (2009). Also called held-out-likelihood, perplexity measures the topic model's ability to predict a test set containing documents that have not been shown to the algorithm during the training phase. Better models would have higher probabilities for these unseen documents. Although the perplexity metric is technically easy to implement it was only used until human judgment measures were proposed by Chang et al. (2009) because it appears to be misleading when it comes to human understanding of topics. The researchers showed a negative correlation between perplexity and human judgment approaches like "word intrusion" and "topic intrusion". This means that, as the perplexity score improves (i.e., the held-out-likelihood is higher), the human interpretability of topics decreases.

Multiple studies attempted to develop quantitative metrics that were positively correlated with the human judgment method. Newman et al. (2010) introduced the notion of topic coherence, proposed the first coherency measure called Pointwise Mutual Information (PMI) and proved that it was a good estimation for the human interpretability of topics. Lau et al. (2014) perform a systematic comparison of different evaluation methods and introduce an improved version of PMI. Normalized PMI (NPMI) reduces the bias of

PMI towards words of lower frequency and provides a standardized range of [1, 1] for the calculated values, where 1 indicated a perfect association. It rescales the probabilities of word co-occurrence and is defined as:

$$NPMI(w_i, w_j) = \frac{\log \frac{P(w_i,w_j)+\epsilon}{P(w_i)\cdot P(w_j)}}{-\log(P(w_i, w_j) + \epsilon)}$$

NPMI applies a negative log co-occurrence probability to each PMI result to normalize them. The final value is computed as the average of all NPMI scores between all top word pairs in a topic. Lau et al. (2014) showed that it correlates better with human judgment than other metrics.

Röder et al. (2015) developed a coherence measure framework, evaluated the most used methods and extended the literature with a new metric, that outperforms existing ones. Among already proposed coherence measures, they find that NPMI (Newman et al., 2010) obtains the best overall results. Their novel metric, called CV, combines the indirect cosine measure with NPMI and the boolean sliding window. According to the experiments of Röder et al. (2015), it achieved the best performance. CV does not use co-occurrence frequency like NPMI, but context vectors. Context vectors count the words that appear within ± five tokens of a given top word. CV restricts this count to only the topic's top words so that vectors share the same length and words. Thus, a context vector is a count of all top topic words seen within ± five tokens of all observations of the given word in all documents. The context vector then computes the NPMI between the given word and the count of top topic words in the vector. For example, given a topic whose top three topics are labeled as a, b, c, the vector for topic a would be:

$$\vec{v_a} = [NPMI(a, a), NPMI(a, b), NPMI(a, c)]$$

Most of the studies researching topic modelling techniques use NPMI (Abuzayed & Al-Khalifa, 2021) (Doan & Hoang, 2021) (Grootendorst, 2022b) or CV (Zhao et al., 2021) (Mifrah & Benlahmar, 2022) or sometimes both (Bennett et al., 2021) for assessing topics coherence.

Apart from metrics trying to capture the coherence of topics, measuring topic diversity is also used for evaluation. This metric is usually defined as the percentage of unique words in all the topics. The range of the metrics is [0, 1], where diversity close to 0 indicates redundant topics and diversity close to 1 implies more varied topics. The more diverse the topics extracted by a model are, the higher will be the coverage of the various aspects of the analyzed corpus. Therefore, researchers aim to develop algorithms that extract topics that are not only coherent but also different from each other (Zhao et al., 2021) (Bennett et al., 2021) (Grootendorst, 2022b).

3 Methodology

As explained in Section 1.1 this thesis aims to answer three research questions regarding topic modelling on alt-tech social media data. First, it examines whether neural topic models perform better than classical ones. Second, it investigates if domain adaptation enhances the performance of neural topic models and which domain adaptation technique should be used. Third, it searches for the best source of alt-tech social media data for adapting topic models on this domain. To provide solid answers seven models tested on the same Parler sample were created. The same testing sample was used because otherwise a fair comparison of the models could not have been possible. In order to adapt the models to the alt-tech domain the following training samples were constructed:

- training sample 1 consisting of 100,000 Parler posts,
- training sample 2 consisting of 100,000 Gab posts and
- training sample 3 consisting of 50,000 Parler posts and 50,000 Gab posts.

3.1 Datasets

The testing and training samples used for experiments are based on two datasets of posts from the well-known alt-tech social media platforms Parler and Gab.

3.1.1 Parler Dataset

As for a previous research work (Bodea, 2022b), the training and testing samples based on Parler data were derived from a large open dataset (Aliapoulios et al., 2021b) consisting of metadata from 13,25 million Parler user profiles, as well as 183 million Parler posts and comments made by 4 million users from August 2018 until January 2021. All information was provided unfiltered in more than 175 newline-delimited JSON (.ndjson) files.

Aliapoulios et al. (2021a) gathered the data and published a paper, providing a first description of the dataset. Their findings reveal that Parler has seen substantial influxes of new members both when celebrities joined the social media platform, and as a reaction to the 2020 U.S. presidential election. They note that the growth of Parler is related to Twitter being more strict with the content of the posts, fact-checking Donald Trump's tweets, and banning him from the platform after the attack on the U.S. Capitol. Until June 2020, the volume of posts and comments remained consistent at about 10,000 per week. However, during the last stage of the 2020 U.S. presidential elections, the number increased to one million posts and comments per week. Moreover, the researchers demonstrate that topics related to conservative views, as well as conspiracy theories and Donald Trump are among the most disputed on Parler. Although only a minority of posts/comments include hashtags, the most common ones were related to the U.S. presidential elections (#trump2020, #maga, and #trump), the alleged election fraud (#stopthesteal, #voterfraud, and #electionfraud) or the QAnon conspiracy theory (#qanon, and #thegreatawakening). This indicates that many of Parler users were Trump supporters debating the 2020 U.S. elections. Another interesting finding reveals that a majority of the content submitted on Parler is received positively by its users,

being 'upvoted' rather than 'downvoted', which leads to the assumption that most users have the same viewpoints.

Due to limited computational power, storage space and time optimization concerns, for the analysis of this thesis four .ndjson files containing parleys were selected. Each .ndjson file consisted of around 110,000 data entries in the form of key-value pairs. The most important keys amongst the 38 stored were: "comments", "body" (containing the text of the post or comment), "bodywithurls", "createdAt" (storing the timestamp the parley was posted at), "createdAtformatted", "upvotes", "downvotes". Because all the data within the dataset was gathered by crawling, the choice of the files should not affect the result in any significant way. This claim is also supported by the congruent results that were achieved with different samples. The selected files from the dataset were the following:

- "parler_data000000000030.ndjson",
- "parler_data000000000031.ndjson",
- "parler_data000000000040.ndjson",
- "parler_data000000000041.ndjson".

Each contains a similar number of records, with the first file having 1,097,921 entries, and the last one 1,095,543.

3.1.2 Gab Dataset

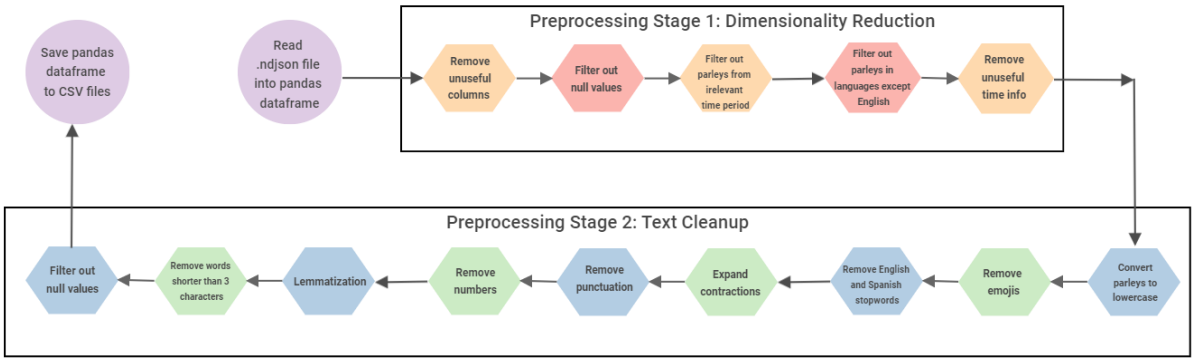
Zannettou et al. (2018a) gathered 22 million posts produced by 336,000 users between August 2016 and January 2018. Only 63% of the posts in the dataset were original posts, the remaining 37% were reposts. The researchers started to analyze the posts by looking at user account creation. They discovered that more users joined the platform in November 2016 and August 2017, when events like the 2016 U.S. presidential elections as well as the Charlottesville "Unite the Right" rally took place. The number of posts increased over time, with around 40% of them written in English. They also looked at widespread hashtags used on the platform and observed they were related to Trump, global news, and politics. Some of the most popular hashtags were "AltRight", "Pizza-gate" and "BanIslam". They indicate that Gab users are followers of the alt-right movement, discussing topics related to conspiracy theories and exchanging Islamophobic opinions. More than 5.4% of all Gab posts include a hate word, which demonstrates that hate speech was used on the platform much more often than on well-known social media networks like Twitter.

For the analysis of this thesis the "gab_posts_jan_2018_030.json" JSON file consisting of around 216,500 gabs was selected.

3.2 Data Preprocessing and Sample Formation

As for most of the NLP tasks, topic modelling also requires text to be preprocessed before training a model. Therefore, in order to help the topic modelling algorithm deliver better results, a thorough selection and cleanup of both the Parler dataset and the Gab dataset was conducted. For this purpose, the Python programming language and the pandas library (*Pandas*, 2022) were used. The preprocessing pipeline followed the steps performed in a previous research (Bodea, 2022b). First, the dimensions of the files selected from the dataset were reduced by selecting only the relevant information. Then, the posts were prepared to be forwarded to the topic modelling algorithms. Each of the four .ndjson files containing parleys and the .json file containing gabs underwent the same preprocessing steps, illustrated in Figure 4.

Figure 4: Preprocessing pipeline



Source: Bodea, A. E. (2022). From freedom of speech to the U.S. Capitol riots: Topic evolution of content on Parler. (Bachelor's Thesis)

The first stage of the preprocessing procedure was selecting relevant information from each file. The first step was reducing the dimension of the file loaded into a pandas dataframe by deleting columns with irrelevant information for the task given. For the Parler files, only two columns were taken into consideration. The first one, labeled as [body], contained the actual text of the parley. The second one, labeled as [createdAtFormatted], consisted of both the date and the time of posting. Every single Parler initial dataframe had almost 1,100,000 rows and exactly 38 columns (key-value pairs of the .ndjson file) and was reduced to only the two relevant columns. For the Gab file, only the textual information stored in the ['post'] column of the dateframe was used. Then, for the dimension reduction of the rows, three operations were performed. First, rows containing null values were filtered out. This resulted in a decrease of the Parler initial dataframe by almost 42%. Second, parleys posted beyond the relevant time period were dropped out from the dataframe, causing a further reduction from around 637,000 to only 353,000 rows for each Parler file. The time period chosen for this research for the parleys was the last stage of the U.S. presidential elections 2020, from November 2020 until January 2021. The last operation for reducing the row dimension was filtering out parleys and gabs whose text contained mostly non-English words. This was made using the fasttext language identification model, which can recognize more than 175 languages (*fastText*,

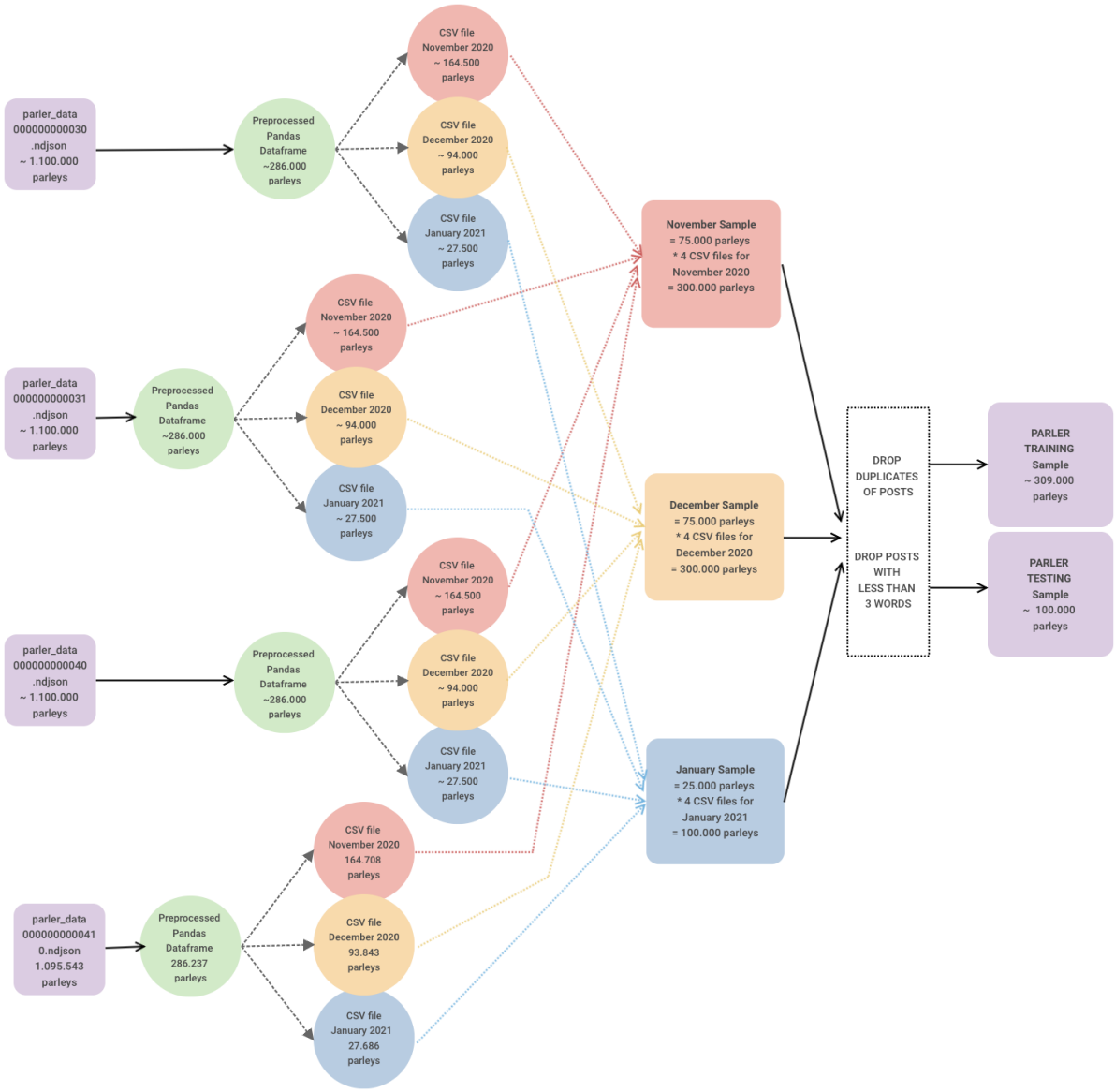
2022). From the column storing the timestamp of the parleys, the part specifying the time was deleted, as only the date was useful for splitting the resulting documents. After the first preprocessing stage, the Gab dataframe contained 1 column and each Parler dataframe 2 columns and around 295,000 posts each.

The second stage of the preprocessing work focused on cleaning the posts and preparing them for the topic modelling algorithm. The first step was simply converting all characters of the posts to lowercase. Then emojis were accurately found and removed from the text. Afterward, the stopwords library (*nltk-stopwords*, 2022) which provides a list of the most common words from a language was imported. Because those words do not hold descriptive power for a topic, both English and Spanish (as the second most common language) stopwords were removed. The next step was expanding contractions commonly found in informal writings like social media posts by using the contractions library. For example, "you're" has been transformed into "you are". Then, all punctuation characters found in the string library have been deleted. The numbers were also removed with the help of the re regular expression module. Afterward, words shorter than three characters were deleted, as the probability for those to provide meaningful information in the topic modelling algorithm was low. Finally, after all the steps refining the posts were completed, filtering out null values was required for a second time. Each resulting Parler dataframe had about 287,000 entries, which represented only a little over 26% of the initial number of parleys. The final Gab dataframe consisted of around 160,000 gabs. The Parler dataframes including only clean posts was divided according to the timestamp into separate dataframes for November 2020, December 2020, and January 2021. All dataframes were saved as CSV files.

As displayed in Figure 5, after preprocessing the four Parler .ndjson files taken into consideration for the analysis, 12 CSV files were produced, four for each of the three months regarded (November, December, January). From these files, four samples were created. The first sample consisted of 300,000 parleys posted in November 2020. It was constructed by choosing a number of 75,000 parleys from each of the four CSV files containing preprocessed parleys from November. The process for the creation of the second sample, December 2020, was the same as for the November 2020 one. For the January 2021 sample, only 100,000 parleys (25,000 from each of the four CSV files containing preprocessed parleys from January) were selected, because Parler was shut down after 10 January 2021. The 3 samples with parleys from a specific month were appended and the duplicates, as well as posts with fewer than 3 words, were dropped out. Then, for the Parler training sample exactly 100,000 posts were randomly selected and the rest of the parleys, more exactly 309,068 posts, were transformed into the testing sample.

The Gab testing sample consisted also of exactly 100,000 posts that were randomly selected from the preprocessed dataframe with 160,041 gabs.

Figure 5: Parler sample formation



Adapted from: Bodea, A. E. (2022). From freedom of speech to the U.S. Capitol riots: Topic evolution of content on Parler. (Bachelor's Thesis)

3.3 Topic Models

As mentioned before, for the experiments conducted in this study seven models were implemented. The code for the models, as well as for preprocessing the datasets and creating the training and testing samples was added to a public GitHub repository (Bodea, 2022a).

The seven models are:

- LDA,
- NMF,
- Model 0: BERTopic with pre-trained SBERT,
- Model 1: BERTopic with SBERT trained using TSDAE on Parler data,
- Model 2: BERTopic with SBERT trained using SimCSE on Parler data,
- Model 3: BERTopic with SBERT trained using TSDAE on Gab data,
- Model 4: BERTopic with SBERT trained using TSDAE on Parler and Gab data.

As explained in the previous section, the models were tested on the same data sample, which enables a fair comparison of their performance. For all seven models, the hyper-parameter specifying the number of topics was set to 10. This number was chosen after several optimization attempts of both the accuracy of the results and the required execution time. Training, as well as testing, was made on Google Colaboratory, which provides fast GPUs. Using a GPU for training neural models shortens the required execution time.

A great number and variety of models were chosen for the experiments in order to provide trustworthy responses to the research questions. The first question examines whether it makes sense to use neural topic models (NTMs), rather than traditional ones for alt-tech social media data. Therefore, both neural and classic models were implemented. Because of their popularity, LDA and NMF were chosen as representatives of traditional topic models. The reason for selecting both models was the different mechanisms they rely on, as explained in Sections 2.2.1 and 2.2.2. For the NTM category, BERTopic was chosen, because research proved it delivers great results (Grootendorst, 2022b) (Mifrah & Benlahmar, 2022) (Egger & Yu, 2022) (Abuzayed & Al-Khalifa, 2021). BERTopic Model 0, which employs the pre-trained SBERT sentence embeddings, is compared to LDA and NMF for answering the first research question. The other experiments conducted for this thesis rely heavily on variations of the BERTopic algorithm. BERTopic Models 1, 2, 3 and 4 were all domain adapted on alt-tech social media data. However, either the way of training or the data used for training differed for each of the models. The first level of diversification was related to the technique used for domain adaptation. BERTopic Model 1 and Model 2 are both trained on Parler data, but Model 1 uses the TSDAE technique, while Model 2 tries out SimCSE for domain adaptation. By comparing Models 1 and 2 with Model 0, research question 2 which investigates the performance of domain adapted algorithms could be answered accurately. The third research question analysis which data source leads to the highest improvement in performance when domain adapting the models. To provide a valid response to this question, Models 1, 3 and 4 are compared.

Each of the three models was domain adapted using the TSDAE technique, but the data used for training was different. The training sample for Model 1 consisted only of parleys, the one for Model 3 only of gabs, while the sample Model 4 was trained on 50% parleys and 50% gabs. As specified before, each training sample consisted of 100,000 posts. This number has been chosen, because Wang et al. (2021), the researchers who introduced the TSDAE technique concluded that the downstream performance usually stagnates when training with more than 100,000 sentences.

3.3.1 LDA and NMF using OCTIS

Representing two of the most well-known traditional topic models, LDA and NMF have been shown to achieve great results and serve as baselines for the comparison of the performance of topic models (Doan & Hoang, 2021). Both LDA and NMF were implemented using OCTIS (Optimizing and Comparing Topic Models is Simple), a framework for training, analyzing, and comparing topic models (Terragni et al., 2021). This framework was also chosen by Grootendorst (2022b) when assessing BERTopic's performance because it provides easily accessible datasets, topic models and evaluation metrics.

3.3.2 BERTopic Models with SBERT

The implementation of the five BERTopic models followed the guidelines provided by Grootendorst (2022a). To prevent any stochastic behavior, a random state was set in the UMAP model of each BERTopic algorithm. This makes the reproduction of the results possible but comes at the expense of performance. Model 0 used the default "all-MiniLM-L6-v2" pre-trained SBERT embeddings, which were proven to work well for English documents. All the other models rely on domain adapted versions of the SBERT sentence embeddings constructed from the 'bert-base-uncased' transformer model. For Models 1, 3 and 4 the TSDAE technique for further training the embeddings was implemented as described by Wang et al. (2022). Model 2 followed the instruction made by Gao et al. (2022).

3.3.3 Evaluation

The evaluation of the topic models used human observation and interpretation, as well as metrics that have been proven to correlate with human judgment. To analyze the results of the models, the 10 topics extracted by each of them were saved in a tabular format. For the BERTopic models some additional figures like a barchart, a hierarchy representation and a map were produced. They represent the results more visually appealing and were added to the GitHub repository alongside the code. In addition to the visual representations, categories were created and each topic was assigned to at least one of them. An analysis of the most frequent words in the topics was also conducted.

Apart from the observation-based evaluation, the human interpretability of topics was tested too. As explained in Section 2.4 the "gold standard" for evaluating topic models is represented by two interpretation tasks developed by Chang et al. (2009), one of them called "word intrusion". Thus, a survey with 25 questions regarding the five BERTopic models was created using Qualtrics, a specialized software for data analysis (Qualtrics,

2022). From each model the first five words from the first five topics along with an "intruder" word have been presented to the respondents. This "intruder" word has been selected from words defining other topics. The respondents were then asked to find the word that did not belong with the others. For example, the question related to the first topic from Model 1 consisted of the five words {vaccine, COVID, mask, virus, flu} and the intruder "light" which was a word extracted for topic 5. In the case of the third topic extracted by Model 0 the question included the following words {party, republican, democrat, RINO, GOP} along with the intruder word "death" selected from topic 8 of Model 0. All questions, as well as the survey description, can be found in the Appendix.

For the evaluation to be as complete as possible, the topics extracted by the models were also assessed using automatic metrics. Both topic diversity and topic coherence, using the NPMI and CV metrics, were implemented through the OCTIS framework (Terragni et al., 2021).

4 Results

4.1 Extracted Topics

The aim of this thesis is to examine whether domain adaptation enhances the performance of topic models. Thus, seven topic modelling algorithms were implemented, their performance was evaluated and a comprehensive comparison was drawn.

The first five words of the 10 topics extracted by each of the seven models are displayed in Table 1. The labels of the topics range from Topic 0 to Topic 9. The first two columns of the table display the topics extracted by LDA and NMF, the traditional topic modelling techniques. The following 5 columns show the results of the different versions of the BERTopic algorithm. As explained in Section 3.3, BERTopic Model 0 relies on the pre-trained SBERT, while the other BERTopic models implement different strategies for domain adaptation. BERTopic Model 1, Model 3 and Model 4 all use SBERT sentence embeddings trained with the TSDAE technique. However, the three models are trained on distinct data samples. Model 1 uses the sample consisting exclusively of Parler posts, Model 3 the one entirely with Gab posts, while Model 4 is trained on the sample with 50% parleys and 50% gabs. Model 2 was build with the other sentence embeddings training technique, SimCSE, and uses the same training sample as Model 1, the one containing exclusively parleys.

In some cases, the first five words in a topic, even though selected by the algorithm as the most relevant, do not hold enough descriptive power to ensure the meaning of the topic was properly understood. Therefore, a separate complete table for each model containing all 10 extracted words for all the 10 topics was added in the Appendix.

To make the comprehension of the results more accessible, methods to analyze and visualize the topics computed by the seven models, as well as evaluation metrics were used. First, the most frequent words occurring in the 700 words (100 words extracted by each topic model) were selected. Then, topic categories were created in order to understand the main discussion subjects on the platform. For the quantitative evaluation of the models, the "gold standard" in form of a survey, as well as statistical metrics are discussed in the following.

	LDA	NMF
Topic 0	law constitution court congress senator	trump supporter suck would donald
Topic 1	not people know get would	you are they time have
Topic 2	election trump state not republican	not doe did will know
Topic 3	are you they going coming	president god biden country america
Topic 4	not would twitter parler video	election vote biden state fraud
Topic 5	like antifa look blm police	need country back take make
Topic 6	god trump president patriot love	going think one right take
Topic 7	biden year trump president last	get not cannot let will
Topic 8	news watch china fake money	people american god know want
Topic 9	supporter big medium trump lol	like would news one look

	BERTopic: Model 0
Topic 0	china chinese ccp virus biden
Topic 1	parler maga follow welcome glad
Topic 2	party republican democrat rinos rino
Topic 3	antifa blm terrorist protest supporter
Topic 4	god pray praying prayer bless
Topic 5	patriot proud boy true thank
Topic 6	georgia recount vote election senate
Topic 7	treason hang dementia tribunal treasonous
Topic 8	racist white black house racism
Topic 9	fox news newsmax watch wallace

	BERTopic: Model 1	BERTopic: Model 2
Topic 0	vaccine covid mask virus not	twitter facebook account parler google
Topic 1	parler twitter follow welcome glad	fox news tucker newsmax watch
Topic 2	fbi barr biden hunter doj	eat cream ice food eating
Topic 3	party republican cruz not senator	god jesus lord praying christ
Topic 4	thank bless god christmas happy	court evidence case supreme lawsuit
Topic 5	god lord jesus amen pray	president joe biden ever never
Topic 6	antifa police blm supporter capitol	texas state red blue california
Topic 7	stopthesteal maga trump wwg electionfraud	jail prison treason need death
Topic 8	fox news tucker newsmax watch	doe not know like matter
Topic 9	nut ball like not hollywood	vaccine virus not covid take

	BERTopic: Model 3	BERTopic: Model 4
Topic 0	biden joe president hunter concede	vaccine covid virus mask not
Topic 1	parler echo welcome glad follow	god jesus lord amen pray
Topic 2	party republican rinos rino not	flynn bless christmas thank happy
Topic 3	twitter facebook account rumble tech	bitch shit like nut lmao
Topic 4	god praying amen jesus lord	party republican rino rinos new
Topic 5	boy basement proud like nut	fox newsmax news tucker watch
Topic 6	china chinese virus biden beijing	ballot vote machine voting recount
Topic 7	not people country reset order	state court supreme vote elector
Topic 8	fox newsmax news watch oan	antifa blm police defund burn
Topic 9	antifa blm police supporter defund	fbi doj barr bill swalwell

Table 1: Topics

4.2 Word Frequency and Topic Categories

Word	Frequency
not	46
trump	18
god	11
would	8
vote	8
people	8
you	8
republican	8
biden	8
supporter	7
news	7
fox	7
like	7
get	7
know	7
state	7
president	6
bless	6
watch	6
thank	6

Table 2: Word frequency

The first step in the analysis of the topics consisted in extracting the most frequent words. As displayed in Table 2 the word with the most occurrences was "not", which is not especially useful for interpreting the meaning of the topics. However, if "not" had been considered a stopword and had been completely deleted from the posts in the preprocessing stage, the meaning would have been altered to a high extent. Understanding if posts express agreement or disagreement is essential. For example, Topic 0 in BERTopic Model 1 is described by the following words {vaccine, covid, mask, virus, not}. The word "not" is required to properly understand the meaning of the topic. It seems like people discussing the COVID-19 global pandemic were rather against restrictions such as wearing a mask and disregarded solutions like the vaccine. The second most repeated word within the topics was "trump", which demonstrates the highly political nature of Parler. This is also supported by other recurrent words like "vote", "republican", "biden", "state" or "president". The word "god" ranks third in the frequency hierarchy with 11 appearances. This proves that most of the models detected discussions related to religious subjects. Other words that occur 6 or 7 times are "news", "fox" and "watch", which suggests that the users of Parler encouraged each other to follow conservative news channels.

For an even more in-depth understanding of the 70 topics, five categories were created and each topic has been assigned to one or, in some cases, two of them. The division of the topics into categories is presented in Table 3, along with an overview of the percentages of topics that fall into each category. Because some topics could not be associated with a single category but with two, the percentages add up to more than 100%, which means they cannot be interpreted as probabilities. The five categories are:

1. Politics
2. Parler
3. Pandemic
4. Religion
5. Others

For almost every model 60% or 70%, so 6 or 7 topics out of 10, were related to politics. While in Model 0 (BERTopic with pre-trained SBERT) all the topics were somehow connected to political events or personalities, for Model 1, which was domain adapted on Parler posts using the TSDAE technique, only 5 out of 10 topics were political. This can be a sign that domain adaptation improves topic extraction because the topics are rather fine-grained, not similar. The second topic category is referring to the Parler alt-tech social media platform itself. Almost all models had a topic including information

	Politics		Parler	
	Topics Nr	Percentage	Topics Nr	Percentage
LDA	0, 2, 5, 6, 7, 8, 9	70%	4	10%
NMF	0, 3, 4, 5, 6, 9	60%	-	0%
Model 0	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	100%	1	10%
Model 1	2, 3, 6, 7, 8	50%	1	10%
Model 2	1, 4, 5, 6, 7, 8	60%	0	10%
Model 3	0, 2, 6, 7, 8, 9	60%	1, 3	20%
Model 4	2, 4, 5, 6, 7, 8, 9	70%	-	0%

	Pandemic		Religion		Others	
	Topics Nr	Percentage	Topics Nr	Percentage	Topics Nr	Percentage
LDA	-	0%	6	10%	1, 3	20%
NMF	-	0%	3, 8	20%	1, 2, 7	30%
Model 0	0	10%	4, 5	20%	-	0%
Model 1	0	10%	4, 5	20%	9	10%
Model 2	9	10%	3	10%	2	10%
Model 3	0, 6	20%	4, 5	20%	5	10%
Model 4	0	10%	1, 2	20%	3	10%

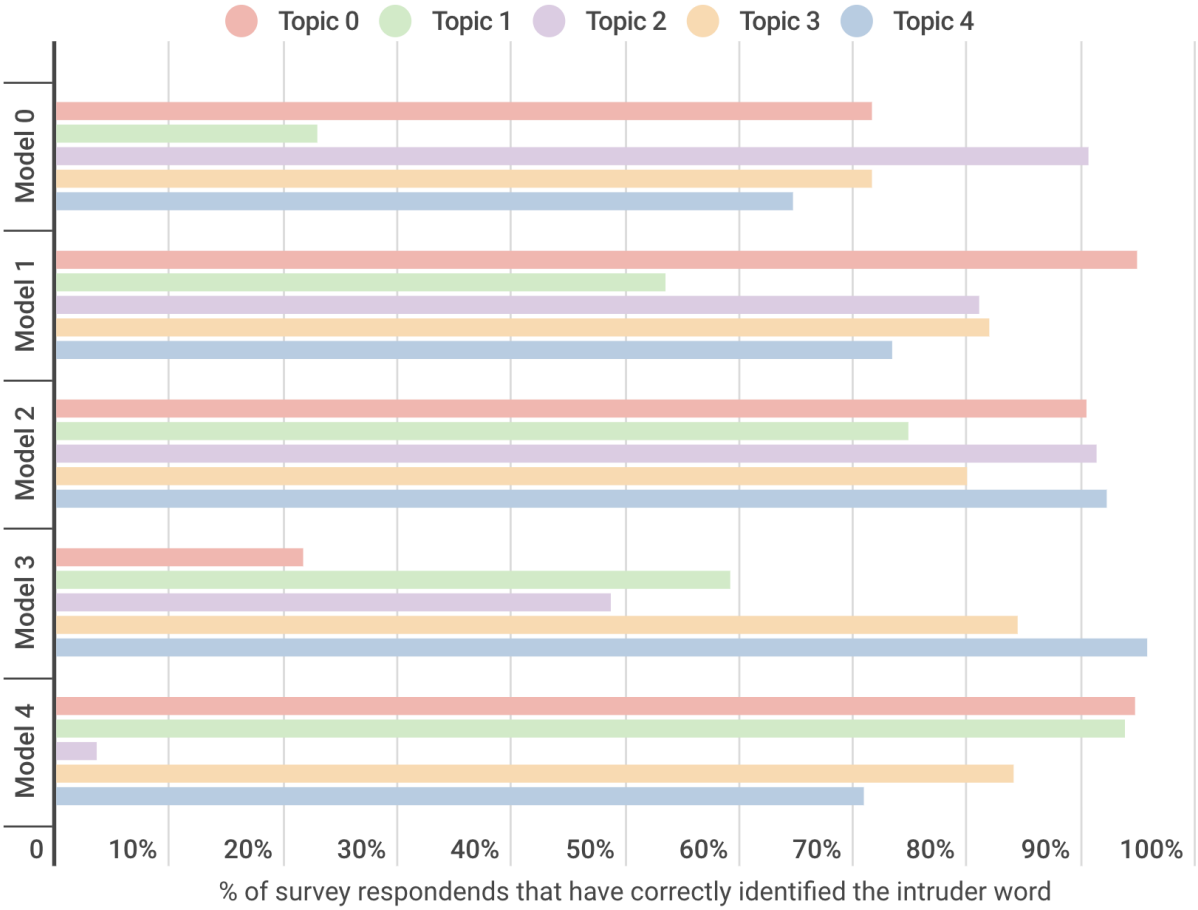
Table 3: Topic categories

about Parler. For example, topic 1 from Model 1 contains the following words {parler, twitter, follow, welcome, glad, account, post, facebook, content, not}. They indicate that people on the platform discussed leaving mainstream social media platforms that restricted the content posted. The third category concerned the COVID-19 pandemic, which was ongoing by the time the parleys from the training and testing samples were posted (November 2020 - January 2021). Apart from the two traditional topic modelling techniques, namely LDA and NMF, all the other models extracted at least one topic connected to the pandemic. For example, topic 9 from Model 2 consists of words clearly related to the COVID-19 crisis: {vaccine, virus, covid, flu}. The fourth category created is religion. In every model, one or even two out of 10 discussion topics are related to this category. For instance, topic 5 from Model 1 consisted of the following words {god, lord, jesus, amen, pray, truth, evil} clearly referring to some religious discussions. Some of the topics could not be assigned to any of the previous four categories because their meaning was ambiguous, so a fifth category labeled as "Others" was created. Among the classic models, two of the topics extracted by LDA and even three computed by NMF were not meaningful, which hints at poor performance. Surprisingly, among the BERTopic versions only Model 0, which uses the pre-trained sentence embeddings, extracted 10 topics out of 10 with an easily intelligible meaning. The other BERTopic models, although domain adapted on alt-tech social media data from Parler or/and Gab, each extracted one topic whose meaning is unclear.

4.3 Survey

As explained in Section 2.4, the most reliable, but at the same time resource-consuming type of topic model evaluation uses human judgment. For this thesis, a survey with 25 questions regarding the five BERTopic models was created according to the guidelines provided by Chang et al. (2009). The survey was filled out by a total of 132 people and an analysis of their responses is displayed in Figure 6. The figure shows the percentage of respondents that have correctly identified the intruder word in each question. For both BERTopic Model 1 and BERTopic Model 2 more than 50% percent of the people found the word that did not belong with the others in each of the five questions. Among the two models, Model 2, which uses SimCSE as the domain adaptation technique, seems to deliver the most easily interpretable results. From Model 0, which was not domain adapted at all, only 23% of people correctly identified the intruder in the second topic, implying that this topic had poor coherence. Even worse, two out of the five topics from Model 3 tested in the survey, were not judged accurately by the people responding to the questions. This is surprising because Model 3 was also domain adapted, but the training data consisted exclusively of Gab posts. Similar to Model 0, one question regarding the topics extracted by Model 4 was answered by less than 5% of people correctly, implying complete lack of coherence.

Figure 6: Survey evaluation



4.4 Quantitative Metrics

To enable a truly in-depth comparison of the models, apart from observation-based and interpretation-based evaluation metrics, automated methods were also used. Topic diversity, as well as topic coherence, was computed. Table 4 displays the results of the metrics for each of the seven topic models. For every evaluation measure, a higher value means that the topics extracted by the model are either more varied or more coherent.

For assessing the coherence of topics, two of the metrics that correlate best with human judgment, namely NPMI and CV, were calculated. The traditional models, LDA and NMF, are outperformed by each version of BERTopic on both of the metrics. Among the BERTopic versions, the one that achieves the highest NPMI and CV scores is Model 1, which was trained with the TSDAE technique using only Parler data. Interestingly, BERTopic Model 0, which was not domain adapted at all, ranks second most coherent model. BERTopic Model 2 achieves the same NPMI coherence as Model 0, but performs worse according to the CV score, although it has been trained on Parler data as well, but using another technique, the SimCSE one.

When analyzing the topic diversity scores, the ranking changes considerably. All the BERTopic models still outperform the classic NMF model, but LDA achieves a better result than BERTopic Model 0. However, the domain adapted BERTopic models have higher diversity scores than LDA. Model 2, the only one using the SimCSE technique, performs worse than Models 1, 3 and 4 which are based on TSDAE. Interestingly, among the domain adapted BERTopic versions using the TSDAE technique, Model 1 has a slightly worse performance than Models 3 and 4. This indicates that not training exclusively on data from the alt-tech social media platform used for testing, offers advantages for the diversity of the results. Overall, the best performance in terms of topic diversity is achieved by BERTopic model 3, which relies on TSDAE and on Gab data for domain adaptation.

	Topic Coherence NPMI	Topic Coherence CV	Topic Diversity
LDA	0.0508	0.58	0.87
NMF	0.0420	0.59	0.69
BERTopic Model 0	0.1269	0.69	0.83
BERTopic Model 1	0.1537	0.71	0.89
BERTopic Model 2	0.1269	0.67	0.88
BERTopic Model 3	0.1203	0.65	0.91
BERTopic Model 4	0.1251	0.68	0.90

Table 4: Topic evaluation metrics

5 Discussion

5.1 Research Questions Answers

This research examined the influence of domain adaptation on the NLP task of topic modelling when dealing with alt-tech social media data. The purpose of this thesis was to provide research-based answers to three questions. As mentioned previously, all the models used in this study were tested on the same data sample, one that consisted of more than 309,000 Parler posts.

The first question took a broader look at the subject and investigated whether classic topic modelling algorithms or neural models achieve better performance on alt-tech social media data. The answer to this question allowed making an informed decision about which model should be domain adapted. For the comparison, two well-known traditional topic models were selected, LDA and NMF, and one neural topic model, BERTopic. As explained before, BERTopic makes use of SBERT sentence embeddings, which are customizable. However, for answering this first question only BERTopic Model 0 is required because only this model uses the pre-trained SBERT. The results of the quantitative evaluation clearly show that NMF is outperformed by BERTopic on all the metrics. LDA achieves worse topic coherence scores calculated with both NPMI and CV metrics, but, in terms of topic diversity, the traditional model seems to perform better than BERTopic. However, when looking at which categories are the extracted topics assigned to, the diversity of LDAs' topics does not necessarily seem higher. All topics resulting from BERTopic Model 0 have a political reference, but that does not mean that the model completely lacks adequate topic isolation capacity. As explained in Section 3, not all topics extracted by all models could be assigned to one single category and this is the case for four topics from Model 0. More precisely, Topic 1 is also associated with the category "Parler", topic 0 with "Pandemic" and topics 4 and 5 include words related to "Religion". Although some topics extracted by BERTopic Model 0 are not referring to a single category, what is more significant is that none of the topics fall under the category "Others". This means that all the topics are easily understandable, which is not the case for the topics extracted by the traditional topic models. For instance, three out of ten topics computed by NMF and two by LDA were assigned to the category "Others" because their meanings are unclear. Moreover, neither NMF nor LDA extracted topics related to the pandemic. This is surprising because they were both tested on the Parler sample containing posts from a period when the pandemic was still ongoing. All the other models found at least one out of the ten topics related to this subject. Even worse, NMF could not find any topics making reference to Parler. These observations prove that the classic topic models were not able to illustrate various subjects. Overall, the chosen neural topic model, BERTopic, outperformed the traditional topic modelling alternatives when dealing with alt-tech social media posts. This conclusion is in line with the findings of previous studies that analyzed the performance of NTMs (Zhao et al., 2021) (Bennett et al., 2021).

Having proven that neural topic models improve topic discovery in comparison with traditional models, domain adaptation could be applied. The second research question investigated whether domain adapted neural topic models achieve greater performance than pre-trained ones. To answer this question two variations of BERTopic Model 0, which uses the pre-trained SBERT model, were implemented. BERTopic Model 1 relies on

a SBERT sentence embeddings model, which was fine-tuned using the TSDAE technique before being forwarded to the actual topic modelling algorithm. The same procedure of domain adaptation was applied to the SBERT sentence embeddings of BERTopic Model 2, but another technique was used, namely SimCSE. However, the fine-tuning sample for both Model 1 and Model 2 remained the same, only the training technique was modified. More exactly, the training sample consisted of 100,000 Parler posts. The three models, BERTopic Model 0, 1 and 2, were evaluated and compared by assigning the topics into categories, computing topic coherence and topic diversity metrics, as well as through the survey. As far as the topic categories are concerned, no significant differences between the domain adapted models could be observed. For both Model 1 and Model 2 one topic out of ten was related to the pandemic, and another one to the Parler platform itself. In Model 1 five topics could be assigned to the "Politics" category and two to the "Religion" category, while in Model 2 six topics were related to "Politics" and only one to "Religion". In comparison with topic Model 0, which was not fine-tuned on alt-tech social media data, all topics extracted by the domain adapted models could easily fall into one single category. This implies that domain adaptation, especially when done on data from the same origin as the testing data, results in improved topic isolation. However, one of the topics extracted by Model 1, as well as one from Model 2 were ambiguous and were, therefore, allocated to the category "Others". The results of the survey, which is considered the most reliable evaluation metric, showed that domain adapted models are superior to their pre-trained counterparts. Every question inspecting the coherence of the topics extracted by BERTopic Model 1 and 2 has been answered by more than 50% of respondents correctly, i.e. people could easily identify the "intruder" word. In contrast, one of the five questions regarding topics extracted from BERTopic Model 0, seemed to lack good coherence as only a little over 20% of people filling out the survey could detect the "intruder" word. Apart from the survey, the coherence of topics was also computed using the NPMI and CV metrics. These show that BERTopic Model 1, which implemented the TSDAE technique for domain adapting the sentence embeddings, achieved the best performance. Model 2, although domain adapted too, only reached the same NPMI score and even a lower CV score as Model 0, the baseline. When examining the topic diversity metric it becomes clear that domain adaptation provides performance advantages, with both Model 1 and 2 outperforming Model 0. Overall, the results of the evaluation demonstrate that domain adaptation improves the models' performance for the alt-tech social media field as for many other domains and tasks (Webersinke et al., 2021), (Nguyen et al., 2020), (Kawintiranon & Singh, 2021). Moreover, the comparison between the two domain adaptation techniques for sentence embeddings, namely TSDAE (Model 1) and SimCSE (Model 2), validates previous findings stating that TSDAE is superior (Wang et al., 2021).

The third research question examines what influence the source of the alt-tech social media data used for domain adaptation has on the performance of the models. The answer to the previous research question reveals that domain adapted neural topic models, in particular the ones trained using the TSDAE technique, achieve high-quality results. Starting from this finding, two more variations of the BERTopic model were implemented. This time, not the domain adaptation technique varied, but the sample the models were trained on. As for BERTopic Model 1, BERTopic Model 3 and 4 all relied on TSDAE for

fine-tuning the SBERT sentence embeddings model. While the training sample for Model 1 consisted exclusively of Parler posts, the one for Model 3 included only posts from Gab. Model 4 was also trained on 100,000 posts as the previous models, but one half originated from Parler and the other half from Gab. Like the previous models, Models 1, 3 and 4 were also evaluated using categories, the survey and automated evaluation metrics. Only small differences between the topic distribution into the five categories could be observed. Two of the ten topics extracted by each of the three models were referring to "Religion". Similarly, one topic from each model was ambiguous and was allocated to the category "Others". Interestingly, Model 3 which was domain adapted using a sample of gabs posted before the outbreak of the COVID-19 global pandemic, extracted two topics related to this category. In comparison, BERTopic Models 1 and 3, both further trained on Parler data gathered during the pandemic, found only one out of 10 topics making reference to the health global crisis. Moreover, Model 3, solely relying on gabs for training, considered 2 topics being related to Parler, while Model 4, which uses the mixed domain adaptation sample found none. The results of the survey enable a more precise comparison of the models. As explained earlier, all questions referring to topics of Model 1 were answered correctly by over 50% of the respondents. This was not the case for Models 3 and 4. The question regarding the third topic extracted by Model 4 was inaccurately answered by less than 5% of the respondents. Even worse, people filling out the survey could not correctly detect the intruder in two of the questions regarding Model 3. The coherence scores of BERTopic Model 1, both trained and tested on Parler data outperformed all of the other models. However, in terms of topic diversity, the highest score was achieved by Model 3, closely followed by Model 4. This suggests that domain adaptating models also using data from another alt-tech platform, not only the one the models are tested on, holds advantages for the diversity of the topics. Usually, more diverse topics are able to capture different elements of the corpus. Yet, in terms of the coherence of the topics fine-tuning and testing topic models on alt-tech social media data from the same platform, clearly delivers the best results. These findings are not contradictory, they just refer to different evaluation metrics. Model 4, trained using both Parler and Gab data illustrates a reasonable compromise for both topic coherence and diversity, as it does not achieve the best scores on either of the metrics, but it is competitive overall. However, if the task allows for slightly poorer performance in terms of topic diversity, a model trained and tested on data coming from the same source, which is the case for Model 1, is more advantageous.

5.2 Strengths and Limitations

To the best of the author's knowledge, this thesis provides the first analysis of the utility and performance of domain adaptation techniques for topic extraction on alt-tech social media data.

One of the great strengths of this study is the data-driven approach, which enables delivering objective evidence. Only datasets gathered by researchers were used, which proves that the analysis is trustworthy and the findings unbiased. Moreover, the experiments consisted not only of classic topic modelling techniques but also of novel, neural ones. Also, the evaluation of the models was exhaustive as the study used observation-

based evaluation methods, metrics based on human judgment, as well as quantitative topic coherence and topic diversity measures.

However, the thesis has its own set of limitations. Although implemented in many variations, one single neural topic model was analyzed, namely BERTopic. Also, the algorithms were tested only on Parler data and the training samples were not particularly diverse. Other relevant alt-tech social media platforms with different user bases could have been included. Regarding the evaluation of the models, the survey, considered the "gold standard" measurement, was only filled out a little over 130 times. Moreover, the selection of the "intruder" word is subjective and could have led to simplifying the task or making it more difficult. Apart from the survey evaluation, the other quantitative metrics can sometimes also be inappropriate. A recent study questioned the positive correlation between human judgment and the NMPI topic coherence metric, especially when dealing with neural topic models like BERTopic (Hoyle et al., 2021). Solely relying on these evaluation metrics can lead to an inaccurate assessment of the models.

6 Conclusion

The aim of this bachelor's thesis is to inspect what impact domain adaptation has on the performance of topic models when dealing with alt-tech social media data.

For this purpose, three research questions have been defined and seven models implemented. Because of their popularity, the alt-tech social media platforms chosen for testing and training the models were Parler and Gab (Lewis, 2020). Both platforms have similar functions to larger social media networks (M. Otala et al., 2021) (Zannettou et al., 2018b), but, in contrast to them, do not restrict the content posted or ban people. Due to the emphasis on free speech, alt-tech platforms attract mostly extremist, alt-right, far-right users. As confirmed by the results of the topic models, the main discussion subjects on Parler are politics, religion and the pandemic, and the expressed viewpoints tend to be conservative and controversial (Bodea, 2022b). The prevalence of hate speech, anti-Semitic comments and discussions related to conspiracy theories is also significant (Mathew et al., 2019). Both Parler and Gab were linked to violent events like the U.S. Capitol attack and the Pittsburgh synagogue shooting, which is why it is highly relevant to analyze these platforms (McIlroy-Young & Anderson, 2019).

The study was based on classic and neural topic models. Neural models have been shown to enhance the performance for topic extraction for multiple tasks and domains (Zhao et al., 2021) (Bennett et al., 2021). Nevertheless, well-known, traditional topic models like LDA and NMF, still serve as a valuable baseline (Doan & Hoang, 2021). Topic models, including neural ones like BERTopic, have been previously tested on alt-tech social media data (Bodea, 2022b). However, these models were never specially designed for this domain. Yet, domain adaptation has been proven to enhance the performance of models for numerous tasks and fields (Gururangan et al., 2020) (Webersinke et al., 2021) (Kawintiranon & Singh, 2021).

Therefore, LDA, NMF and several variations of BERTopic have been implemented, thoroughly evaluated and compared. Domain adaptation was performed by further training SBERT sentence embeddings using either the TSDAE or the SimCSE technique. The experiments conducted in this thesis revealed three findings. First, neural topic models outperform traditional models like LDA and NMF when applied to alt-tech social media posts. Second, and maybe the most significant, domain adaptation for neural topic models enhances the performance of topic extraction to a great extent. Third, when domain-adapting neural topic models, the data source used for training influences topic coherence and topic diversity. More precisely, when the training and testing data come from the same alt-tech social media platform, the model achieves particularly high topic coherence scores. However, using data from different platforms results in improved diversity of topics. A balance of both coherence and diversity is reached when models are trained on data gathered from more alt-tech platforms, including the one the model will be tested on.

A Appendix

As explained in the Results section, each of the 10 topics extracted by each of the 7 models was described using 10 words. In some cases, the meaning of the topic can only be properly understood after analyzing all the 10 words of a topic, not just the first five presented in Table 1. Therefore, Tables 5, 6, 7, 8, 9, 10, 11 show the complete descriptions of the topics extracted by each of the models.

One of the evaluation methods used in this thesis was a survey, which contained 25 questions, five for each BERTopic model. All the content of the survey, including the introduction, instructions and the questions shown to the participants can be found below. For each question, the "intruder" is the last word and was marked appropriately. Before being displayed to the participants, the responses to each question were randomized.

Hi!

I am studying different natural language processing algorithms that extract topics from large collections of documents. I trained and tested several models on datasets of posts from the alt-tech social media platforms Parler and Gab. These social networks promote free speech, moderate content less stringently, and thus, attract alt-right, far-right users with extreme or fringe opinions. The posts were gathered from the last phase of the 2020 U.S. presidential election, resulting in Joe Biden, a member of the Democratic party, defeating the former Republican president Donald Trump. In order to evaluate whether the topics have semantic coherence, I kindly ask you to assess how meaningful some of the resulting topics are. Please answer all the questions, it only takes 3 to 5 minutes. All responses will be kept confidential. If you have any questions feel free to contact me. Thank you for your time.

Instructions: The survey consists of 25 questions in the same format. For each question, you will be provided with a set of 6 words, 5 words selected from a topic extracted by one of the algorithms and one random word. Your task is to find the word that does not belong with the ones from the topic, i.e. the intruder.

Example: In the set dog, cat, horse, apple, pig, cow the intruder word is apple because the remaining words dog, cat, horse, pig, cow make sense together - they are all animals.

Disclaimer: Topics were used unfiltered, some questions may contain strong language.

Student: Andreea Elena Bodea
Bachelor's Thesis: Domain adapted topic modelling for alt-tech social media
Institute for Artificial Intelligence in Management
Ludwig Maximilian University of Munich

1. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 0 - Topic 0]
   - ○ China
   - ○ CCP (= Chinese Communist Party)
   - ○ virus
   - ○ Biden
   - ○ communist
   - ● hole
2. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 0 - Topic 1]
   - ○ Parler (referring to: Parler, social media platform)
   - ○ MAGA (= Make America Great Again)
   - ○ follow
   - ○ welcome
   - ○ content
   - ● racism
3. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 0 - Topic 2]
   - ○ party
   - ○ republican
   - ○ democrat
   - ○ RINO (= Republican in Name Only)
   - ○ GOP (= Grand Old Party" = Republican Party)
   - ● death
4. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 0 - Topic 3]
   - ○ Antifa (= anti-fascist political movement)
   - ○ BLM (= Black Lives Matter)
   - ○ terrorist
   - ○ protest
   - ○ supporter
   - ● bless
5. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 0 - Topic 4]
   - ○ god
   - ○ pray
   - ○ bless
   - ○ president
   - ○ Trump
   - ● watching

6. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 1 - Topic 0]
   - ○ vaccine
   - ○ COVID (= Coronavirus Disease)
   - ○ mask
   - ○ virus
   - ○ flu
   - ● light
7. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 1 - Topic 1]
   - ○ Parler (referring to: Parler, social media platform)
   - ○ Twitter
   - ○ follow
   - ○ welcome
   - ○ account
   - ● Antifa (= anti-fascist political movement)
8. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 1 - Topic 2]
   - ○ FBI (= Federal Bureau of Investigation)
   - ○ Barr (referring to: William Barr, member of the Republican Party, U.S. Attorney General in the administration of President Donald Trump)
   - ○ Biden
   - ○ Hunter (referring to: Hunter Biden, American attorney and second son of U.S. President Joe Biden)
   - ○ DOJ (= United States Department of Justice)
   - ● network
9. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 1 - Topic 3]
   - ○ party
   - ○ republican
   - ○ Cruz (referring to: Ted Cruz, member of the Republican Party, U.S. senator for Texas)
   - ○ senator
   - ○ Trump
   - ● peaceful
10. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 1 - Topic 4]
    - ○ thank
    - ○ bless
    - ○ god
    - ○ Christmas
    - ○ happy
    - ● election

11. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 2 - Topic 0]
    - ○ Twitter
    - ○ Facebook
    - ○ account
    - ○ Parler (referring to: Parler, social media platform)
    - ○ Google
    - ● history
12. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 2 - Topic 1]
    - ○ Fox (referring to: The Fox News Channel, American conservative television channel)
    - ○ news
    - ○ Tucker (referring to: Tucker Carlson, conservative political commentator and American television host on FoxNews)
    - ○ Newsmax (referring to: Newsmax TV, American conservative television channel)
    - ○ watch
    - ● state
13. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 2 - Topic 2]
    - ○ eat
    - ○ cream
    - ○ ice
    - ○ food
    - ○ turkey
    - ● anything
14. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 2 - Topic 3]
    - ○ god
    - ○ Jesus
    - ○ Lord
    - ○ praying
    - ○ Christ
    - ● death
15. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 2 - Topic 4]
    - ○ court
    - ○ evidence
    - ○ case
    - ○ supreme
    - ○ lawsuit
    - ● vaccine

16. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 3 - Topic 0]
    - ○ Biden
    - ○ Joe
    - ○ president
    - ○ Hunter (referring to: Hunter Biden, American attorney and second son of U.S. President Joe Biden)
    - ○ concede
    - ● people
17. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 3 - Topic 1]
    - ○ Parler (referring to: Parler, social media platform)
    - ○ echo
    - ○ welcome
    - ○ glad
    - ○ follow
    - ● Beijing
18. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 3 - Topic 2]
    - ○ party
    - ○ republican
    - ○ RINO (= Republican in Name Only)
    - ○ democrat
    - ○ Kemp (referring to: Brian Kemp, member of the Republican Party, governor of Georgia)
    - ● Newsmax (referring to: Newsmax TV, American conservative television channel)
19. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 3 - Topic 3]
    - ○ Twitter
    - ○ Facebook
    - ○ account
    - ○ Rumble (referring to: Rumble, online video platform)
    - ○ tech
    - ● communist
20. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 3 - Topic 4]
    - ○ god
    - ○ praying
    - ○ amen
    - ○ Jesus
    - ○ Lord
    - ● little

21. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 4 - Topic 0]

    - ○ vaccine
    - ○ COVID (referring to: Coronavirus Disease)
    - ○ virus
    - ○ mask
    - ○ flu
    - ● watching

22. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 4 - Topic 1]

    - ○ god
    - ○ Jesus
    - ○ Lord
    - ○ amen
    - ○ pray
    - ● ballot

23. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 4 - Topic 2]

    - ○ Flynn (referring to: Michael Flynn, member of the Republican Party, National Security Advisor in the administration of U.S. President Donald Trump)
    - ○ bless
    - ○ Christmas
    - ○ thank
    - ○ happy
    - ● supporter

24. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 4 - Topic 3]

    - ○ bitch
    - ○ shit
    - ○ like
    - ○ nut
    - ○ lmao (= laughing my ass off)
    - ● FBI (= Federal Bureau of Investigation)

25. For the following set of words please select the word that does not belong with the others, i.e., the intruder. [Model 4 - Topic 4]

    - ○ party
    - ○ republican
    - ○ RINO (= Republican in Name Only)
    - ○ Trump
    - ○ GOP (= Grand Old Party" = Republican Party)
    - ● court

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	law	constitution	court	congress	senator	capitol	corrupt	military	justice	system
Topic 2	not	people	know	get	would	going	want	think	take	one
Topic 3	election	trump	state	not	republican	vote	fraud	party	need	democrat
Topic 4	are	you	they	going	coming	will	not	get	wait	black
Topic 5	not	would	twitter	parler	video	have	get	account	post	see
Topic 6	like	antifa	look	blm	police	war	shit	woman	name	fuck
Topic 7	god	trump	president	patriot	love	you	thank	penny	great	america
Topic 8	biden	year	trump	president	last	joe	communist	vote	obama	ago
Topic 9	news	watch	china	fake	money	fox	paid	traitor	coward	commie
Topic 10	supporter	big	medium	trump	lol	follow	maga	tech	part	social

Table 5: Topics extracted by LDA model

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	trump	supporter	suck	would	donald	maga	win	know	republican	team
Topic 2	you	are	they	time	have	back	know	that	thank	will
Topic 3	not	doe	did	will	know	even	would	want	say	anything
Topic 4	president	god	biden	country	america	never	trump	love	bless	year
Topic 5	election	vote	biden	state	fraud	democrat	ballot	republican	voter	voting
Topic 6	need	country	back	take	make	would	state	stop	stand	time
Topic 7	going	think	one	right	take	know	that	cannot	democrat	would
Topic 8	get	not	cannot	let	will	back	away	time	way	rid
Topic 9	people	american	god	know	want	country	many	america	government	right
Topic 10	like	would	news	one	look	fox	see	good	never	sound

Table 6: Topics extracted by NMF model

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	china	chinese	ccp	virus	biden	not	communist	russia	beijing	country
Topic 2	parler	maga	follow	welcome	glad	content	trump	supporter	you	truly
Topic 3	party	republican	democrat	rinos	rino	dems	gop	coward	not	need
Topic 4	antifa	blm	terrorist	protest	supporter	not	trump	peaceful	people	riot
Topic 5	god	pray	praying	prayer	bless	president	trump	you	thank	amen
Topic 6	patriot	proud	boy	TRUE	thank	american	you	not	god	bless
Topic 7	georgia	recount	vote	election	senate	not	ballot	state	republican	runoff
Topic 8	treason	hang	dementia	tribunal	treasonous	firing	squad	military	death	not
Topic 9	racist	white	black	house	racism	color	race	not	people	matter
Topic 10	fox	news	newsmax	watch	wallace	chris	watching	not	oan	hole

Table 7: Topics extracted by Model 0: BERTopic with pre-trained SBERT

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	vaccine	covid	mask	virus	not	flu	people	death	get	test
Topic 2	parler	twitter	follow	welcome	glad	account	post	facebook	content	not
Topic 3	fbi	barr	biden	hunter	doj	flynn	not	epstein	cia	durham
Topic 4	party	republican	cruz	not	senator	need	trump	gop	senate	vote
Topic 5	thank	bless	god	christmas	happy	dan	you	well	keep	merry
Topic 6	god	lord	jesus	amen	pray	truth	evil	not	prevail	light
Topic 7	antifa	police	blm	supporter	capitol	protest	peaceful	capital	trump	not
Topic 8	stopthesteal	maga	trump	wwg	electionfraud	wga	kag	voterfraud	election	trumptrain
Topic 9	fox	news	tucker	newsmax	watch	hannity	watching	cnn	network	not
Topic 10	nut	ball	like	not	hollywood	you	bitch	suck	retard	face

Table 8: Topics extracted by Model 1: BERTopic with SBERT trained using TSDAE on Parler data

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	twitter	facebook	account	parler	google	deleted	app	not	banned	apple
Topic 2	fox	news	tucker	newsmax	watch	cnn	bye	hannity	network	not
Topic 3	eat	cream	ice	food	eating	turkey	like	not	coffee	drink
Topic 4	god	jesus	lord	praying	christ	pray	bless	amen	light	prayer
Topic 5	court	evidence	case	supreme	lawsuit	not	judge	scotus	state	texas
Topic 6	president	joe	biden	ever	never	trump	basement	history	not	best
Topic 7	texas	state	red	blue	california	florida	color	county	move	not
Topic 8	jail	prison	treason	need	death	put	time	penalty	sentence	not
Topic 9	doe	not	know	like	matter	care	want	even	trump	anything
Topic 10	vaccine	virus	not	covid	take	flu	get	people	speed	would

Table 9: Topics extracted by Model 2: BERTopic with SBERT trained using SimCSE on Parler data

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	biden	joe	president	hunter	concede	not	kamala	trump	harris	sleepy
Topic 2	parler	echo	welcome	glad	follow	post	massupvote	content	maga	truly
Topic 3	party	republican	rinos	rino	not	democrat	mitch	kemp	cruz	new
Topic 4	twitter	facebook	account	rumble	tech	not	duck	big	parlor	deleted
Topic 5	god	praying	amen	jesus	lord	pray	prevail	not	evil	prayer
Topic 6	boy	basement	proud	like	nut	not	bitch	little	back	fuck
Topic 7	china	chinese	virus	biden	beijing	swalwell	fang	communist	spy	not
Topic 8	not	people	country	reset	order	world	ppl	america	american	one
Topic 9	fox	newsmax	news	watch	oan	tucker	watching	network	cnn	oann
Topic 10	antifa	blm	police	supporter	defund	peaceful	not	domestic	trump	protest

Table 10: Topics extracted by Model 3: BERTopic with SBERT trained using TSDAE on Gab data

	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
Topic 1	vaccine	covid	virus	mask	not	flu	people	get	fauci	death
Topic 2	god	jesus	lord	amen	pray	not	evil	prevail	christ	faith
Topic 3	flynn	bless	christmas	thank	happy	god	general	merry	family	pardon
Topic 4	bitch	shit	like	nut	lmao	fucking	face	not	piece	ankle
Topic 5	party	republican	rino	rinos	new	trump	not	gop	patriot	vote
Topic 6	fox	newsmax	news	tucker	watch	hannity	bye	oan	watching	network
Topic 7	ballot	vote	machine	voting	recount	dead	dominion	not	audit	count
Topic 8	state	court	supreme	vote	elector	scotus	penny	election	mike	not
Topic 9	antifa	blm	police	defund	burn	capital	supporter	not	capitol	trump
Topic 10	fbi	doj	barr	bill	swalwell	cia	russia	gate	wray	fang

Table 11: Topics extracted by Model 4: BERTopic with SBERT trained using TSDAE on Parler and Gab data

B List of Abbreviations

$BERT$ Bidirectional Encoder Representations from Transformers

$BPTM$ Bayesian Probabilistic Topic Model

$HDBSCAN$ Hierarchical Density-Based Spatial Clustering of Applications with Noise

LDA Latent Dirichlet Allocation

NLP Natural Language Processing

NMF Non-Negative Matrix Factorization

$NPMI$ Normalized Pointwise Mutual Information

NTM Neural Topic Model

$OCTIS$ Optimizing and Comparing Topic Models is Simple

PMI Pointwise Mutual Information

$SBERT$ Sentence-BERT

$SimCSE$ Simple Contrastive Learning of Sentence Embeddings

$TF - IDF$ Term Frequency – Inverse Document Frequency

$TSDAE$ Transformer-based Sequential Denoising Auto-Encoder

$U.S.$ United States

$UMAP$ Uniform Manifold Approximation and Projection

References

Abuzayed, A., & Al-Khalifa, H. (2021). Bert for arabic topic modeling: an experimental study on bertopic technique. *Procedia Computer Science*, *189*, 191–194.

Albalawi, R., Yeap, T. H., & Benyoucef, M. (2020). Using topic modeling methods for short-text data: A comparative analysis. *Frontiers in Artificial Intelligence*, *3*, 42.

Aliapoulios, M., Bevensee, E., Blackburn, J., Bradlyn, B., De Cristofaro, E., Stringhini, G., & Zannettou, S. (2021a). A large open dataset from the parler social network. In *Icwsm* (pp. 943–951).

Aliapoulios, M., Bevensee, E., Blackburn, J., Bradlyn, B., De Cristofaro, E., Stringhini, G., & Zannettou, S. (2021b, January). *A large open dataset from the parler social network.* Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.4442460` doi: 10.5281/zenodo.4442460

Alt-tech. (2022, Oct). Wikimedia Foundation. Retrieved from `https://en.wikipedia.org/wiki/Alt-tech`

Araci, D. (2019). Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.

Bennett, A., Misra, D., & Than, N. (2021). Have you tried neural topic models? comparative analysis of neural and non-neural topic models with application to covid-19 twitter data. *arXiv preprint arXiv:2105.10165*.

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of machine Learning research*, *3*, 993–1022.

Bodea, A. E. (2022a, 11). *Domain adapted topic modelling for alt-tech social media.* Retrieved from `https://github.com/andreea-bodea/bachelors-thesis-informatics`

Bodea, A. E. (2022b). *From freedom of speech to the U.S. Capitol riots: Topic evolution of content on Parler.* (Bachelor's Thesis). Ludwig-Maximilians-Universität München, Institute of Artificial Intelligence (AI) in Management, Munich, Germany.

Boyd-Graber, J., Hu, Y., Mimno, D., et al. (2017). Applications of topic models. *Foundations and Trends® in Information Retrieval*, *11*(2-3), 143–296.

Chang, J., Gerrish, S., Wang, C., Boyd-Graber, J., & Blei, D. (2009). Reading tea leaves: How humans interpret topic models. *Advances in neural information processing systems*, *22*.

Cheng, X., Yan, X., Lan, Y., & Guo, J. (2014). Btm: Topic modeling over short texts. *IEEE Transactions on Knowledge and Data Engineering*, *26*(12), 2928–2941.

Chuang, J., Manning, C. D., & Heer, J. (2012). Termite: Visualization techniques for assessing textual topic models. In *Proceedings of the international working conference on advanced visual interfaces* (pp. 74–77).

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

Doan, T.-N., & Hoang, T.-A. (2021). Benchmarking neural topic models: An empirical study. In *Findings of the association for computational linguistics: Acl-ijcnlp 2021* (pp. 4363–4368).

Egger, R., & Yu, J. (2022). A topic modeling comparison between lda, nmf, top2vec, and bertopic to demystify twitter posts. *Frontiers in Sociology*, *7*.

fasttext. (2022). Retrieved from `https://fasttext.cc/docs/en/language-identification.html`

Gab. (2022). Retrieved from `https://gab.com/`

Gao, T., Yao, X., & Chen, D. (2021). Simcse: Simple contrastive learning of sentence embeddings. *arXiv preprint arXiv:2104.08821*.

Gao, T., Yao, X., & Chen, D. (2022). *Simcse.* Retrieved from `https://www.sbert.net/examples/unsupervised_learning/SimCSE/README.html`

Grootendorst, M. (2022a). *Bertopic.* Retrieved from `https://github.com/MaartenGr/BERTopic`

Grootendorst, M. (2022b). Bertopic: Neural topic modeling with a class-based tf-idf procedure. *arXiv preprint arXiv:2203.05794*.

Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don't stop pretraining: adapt language models to domains and tasks. *arXiv preprint arXiv:2004.10964*.

Hadiat, A. R. (2022). *Topic modeling evaluations: The relationship between coherency and accuracy* (Master's Thesis). University of Groningen.

Hoyle, A., Goel, P., Hian-Cheong, A., Peskov, D., Boyd-Graber, J., & Resnik, P. (2021). Is automated topic model evaluation broken? the incoherence of coherence. *Advances in Neural Information Processing Systems*, *34*, 2018–2033.

Kawintiranon, K., & Singh, L. (2021). Knowledge enhanced masked language model for stance detection. In *Proceedings of the 2021 conference of the north american chapter of the association for computational linguistics: human language technologies.*

Lau, J. H., Newman, D., & Baldwin, T. (2014). Machine reading tea leaves: Automatically evaluating topic coherence and topic model quality. In *Proceedings of the 14th conference of the european chapter of the association for computational linguistics* (pp. 530–539).

Lee, D., & Seung, H. S. (2000). Algorithms for non-negative matrix factorization. *Advances in neural information processing systems*, *13*.

Lewis, B. (2020, Nov). *The top 5 best alternative social media apps: An alt-tech guide.* Retrieved from https://medium.com/simplylocal/the-top-5-best-alternative-social-media-apps-an-alt-tech-guide-cc4a1c07a412

Li, C., Wang, H., Zhang, Z., Sun, A., & Ma, Z. (2016). Topic modeling for short texts with auxiliary word embeddings. In *Proceedings of the 39th international acm sigir conference on research and development in information retrieval* (pp. 165–174).

MacGillis, A. (2021). *Inside the capitol riot: What the parler videos reveal.* Retrieved from https://www.propublica.org/article/inside-the-capitol-riot-what-the-parler-videos-reveal

Mathew, B., Dutt, R., Goyal, P., & Mukherjee, A. (2019). Spread of hate speech in online social media. In *Proceedings of the 10th acm conference on web science* (pp. 173–182).

McIlroy-Young, R., & Anderson, A. (2019). From "welcome new gabbers" to the pittsburgh synagogue shooting: The evolution of gab. In *Proceedings of the international aaai conference on web and social media* (Vol. 13, pp. 651–654).

McInnes, L., Healy, J., & Astels, S. (2017). hdbscan: Hierarchical density based clustering. *J. Open Source Softw.*, *2*(11), 205.

McInnes, L., Healy, J., & Melville, J. (2018). Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.

Mifrah, S., & Benlahmar, E. H. (2022). Topic modeling with transformers for sentence-level using coronavirus corpus. *International Journal of Interactive Mobile Technologies*, *16*(17).

M. Otala, J., Kurtic, G., Grasso, I., Liu, Y., Matthews, J., & Madraki, G. (2021). Political polarization and platform migration: a study of parler and twitter usage by united states of america congress members. In *Companion proceedings of the web conference 2021* (pp. 224–231).

Newman, D., Lau, J. H., Grieser, K., & Baldwin, T. (2010). Automatic evaluation of topic coherence. In *Human language technologies: The 2010 annual conference of the north american chapter of the association for computational linguistics* (pp. 100–108).

Nguyen, D. Q., Vu, T., & Nguyen, A. T. (2020). Bertweet: A pre-trained language model for english tweets. *arXiv preprint arXiv:2005.10200*.

nltk-stopwords. (2022). Retrieved from https://openbase.com/js/nltk-stopwords/documentation

Pandas. (2022). Retrieved from https://pandas.pydata.org/

Parler. (2022). Retrieved from https://parler.com/

Parler community guidelines. (2021). Retrieved from https://www.parler.com/documents/guidelines.pdf

Prabhu, A., Guhathakurta, D., Subramanian, M., Reddy, M., Sehgal, S., Karandikar, T., ... others (2021). Capitol (pat) riots: A comparative study of twitter and parler. *arXiv preprint arXiv:2101.06914*.

Qiang, J., Qian, Z., Li, Y., Yuan, Y., & Wu, X. (2020). Short text topic modeling techniques, applications, and performance: a survey. *IEEE Transactions on Knowledge and Data Engineering*, *34*(3), 1427–1445.

Qualtrics. (2022). *Qualtrics survey maker.* Retrieved from https://www.qualtrics.com/uk/free-account/

Reimers, N., & Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*. Retrieved from https://arxiv.org/abs/1908.10084 doi: 10.48550/ARXIV.1908.10084

Röder, M., Both, A., & Hinneburg, A. (2015). Exploring the space of topic coherence measures. In *Proceedings of the eighth acm international conference on web search and data mining* (pp. 399–408).

Shi, T., Kang, K., Choo, J., & Reddy, C. K. (2018). Short-text topic modeling via non-negative matrix factorization enriched with local word-context correlations. In *Proceedings of the 2018 world wide web conference* (pp. 1105–1114).

Terragni, S., Fersini, E., Galuzzi, B. G., Tropeano, P., & Candelieri, A. (2021, apr). OCTIS: Comparing and optimizing topic models is simple! In *Proceedings of the 16th conference of the european chapter of the association for computational linguistics: System demonstrations* (pp. 263–270). Association for Computational Linguistics. Retrieved from https://aclanthology.org/2021.eacl-demos.31https://github.com/MIND-Lab/OCTIS doi: 10.18653/v1/2021.eacl-demos.31

Toetzke, M., Banholzer, N., & Feuerriegel, S. (2022). Monitoring global development aid with machine learning. *Nature Sustainability*, 1–9.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, *30*.

Wallach, H. M., Murray, I., Salakhutdinov, R., & Mimno, D. (2009). Evaluation methods for topic models. In *Proceedings of the 26th annual international conference on machine learning* (pp. 1105–1112).

Wang, K., Reimers, N., & Gurevych, I. (2021). Tsdae: Using transformer-based sequential denoising auto-encoder for unsupervised sentence embedding learning. *arXiv preprint arXiv:2104.06979*.

Wang, K., Reimers, N., & Gurevych, I. (2022). *Tsdae.* Retrieved from https://www.sbert.net/examples/unsupervised_learning/TSDAE/README.html

Webersinke, N., Kraus, M., Bingler, J. A., & Leippold, M. (2021). Climatebert: A pre-trained language model for climate-related text. *arXiv preprint arXiv:2110.12010*.

Yin, J., & Wang, J. (2014). A dirichlet multinomial mixture model-based approach for short text clustering. In *Proceedings of the 20th acm sigkdd international conference on knowledge discovery and data mining* (pp. 233–242).

Zannettou, S., Bradlyn, B., De Cristofaro, E., Kwak, H., Sirivianos, M., Stringini, G., & Blackburn, J. (2018a, September). *Dataset for "what is gab? a bastion of free speech or an alt-right echo chamber?"*. Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.3460400` doi: 10.5281/zenodo.3460400

Zannettou, S., Bradlyn, B., De Cristofaro, E., Kwak, H., Sirivianos, M., Stringini, G., & Blackburn, J. (2018b). What is gab: A bastion of free speech or an alt-right echo chamber. In *Companion proceedings of the the web conference 2018* (pp. 1007–1014).

Zhao, H., Phung, D., Huynh, V., Jin, Y., Du, L., & Buntine, W. (2021). Topic modelling meets deep neural networks: A survey. *arXiv preprint arXiv:2103.00498*.